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Technical Specification

R1 interface: Test Specification

O-RAN ALLIANCE e.V., Buschkauler Weg 27, 53347 Alfter, Germany

Register of Associations, Bonn VR 11238, VAT ID DE321720189

Contents

Foreword

This Technical Specification (TS) has been produced by WG2 of the O-RAN ALLIANCE. It is part of a TS-family covering the WG2: R1 Interface Specifications.

The content of the present document is subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN ALLIANCE modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

version xx.yy.zz

where:

xx: the first digit-group is incremented for all changes of substance, i.e., technical enhancements, corrections, updates, etc. (the initial approved document will have xx=01). Always 2 digits with leading zero if needed.

yy: the second digit-group is incremented when editorial only changes have been incorporated in the document. Always 2 digits with leading zero if needed.

zz: the third digit-group included only in working versions of the document indicating incremental changes during the editing process. External versions never include the third digit-group. Always 2 digits with leading zero if needed.

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

Introduction

The purpose of the present document is to specify conformance and interoperability test cases for the R1 interface. Test cases and methodology for the R1 Service Consumer and R1 Service Producer, are separately specified.

The present document contains R1 test cases for conformance testing and interoperability testing. For each specified test case, cover the entrance criteria, procedure and expected result. The test cases are defined based on the procedures, interface and functional requirements as specified in the R1 specifications series for which the test cases are used to validate conformance and interoperability.

1 Scope

The present document specifies test cases for conformance testing and interoperability testing of the rApps and R1 services over R1 interface.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

[1] O-RAN.WG2.TS.R1GAP: "R1 General Aspects and Principles"("R1GAP").

[2] O-RAN.WG2.TS.R1UCR: "R1 Use Cases and Requirements"("R1UCR").

[3] O-RAN.WG2.TS.R1TP: "Transport Protocols for R1 services"("R1TP").

[4] O-RAN.WG2.TS.R1AP: "Application Protocols for R1 services"("R1AP").

[5] O-RAN.WG2.TS.Non-RT-RIC-ARCH: "Non-RT RIC: Architecture"("Non-RT RIC ARCH").

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are not necessary for the application of the present document, but they assist the user with regard to a particular subject area.

[i.1] 3GPP TR 21.905: “Vocabulary for 3GPP Specifications”.

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in R1GAP [1], R1UCR [2], R1TP [3] apply.

3.2 Symbols

Void

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in R1GAP [1], R1UCR [2], R1TP [3] apply.

4 Test methodology

4.1 General

This clause describes the methodology for conformance and interoperability testing of rApps and SMO/Non-Real Time RIC framework over R1 interface.

For conformance tests, simulators are used for testing R1 procedures. These simulators will have capability of generating HTTP requests and responses. There will be flexibility in configuring URI, headers and body for these HTTP requests and responses to enable creation of various test cases.

For interoperability tests, devices under tests are rApps and SMO/Non-RT RIC framework that are defined in the Non-RT RIC architecture specification [5], these devices are brought to operation by connecting to appropriate real or simulated devices.

4.2 Conformance testing of rApps

4.2.1 General

For conformance testing of rApps, rApp is the device under test and SMO/Non-RT RIC framework is the test simulator.

The present document specifies conformance tests for API Consumer and API Producer functionality as specified in R1AP [4].

4.2.2 Test configuration

4.2.2.1 Overview

The test configuration for R1 conformance testing of rApp is illustrated in figure 4.2.2.1-1. For testing of rApp over R1 interface, the rApp is onboarded and instantiated in the cloud environment of the test simulator.

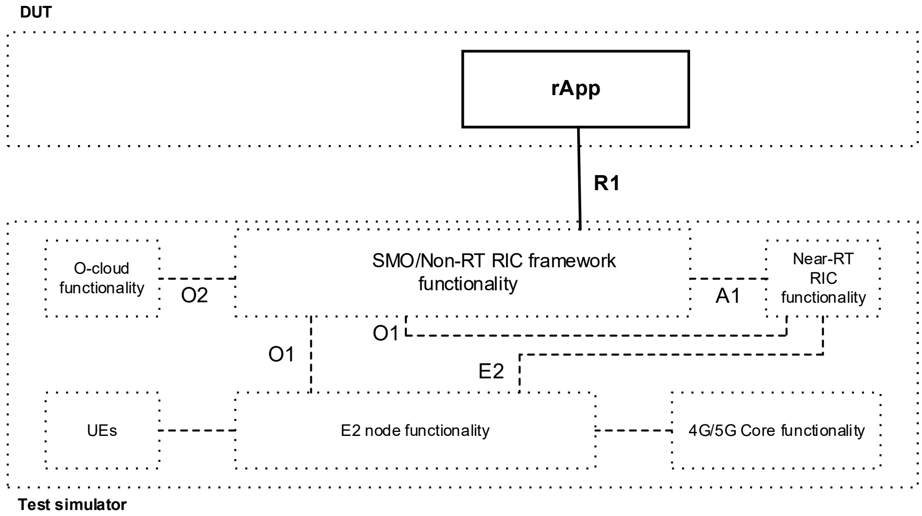

Figure 4.2.2.1-1: Illustration of R1 conformance testing of rApps.

4.2.2.2 Device under test rApps

For enabling conformance testing, rApp has implemented API Consumer and/or API Producer functionality and the procedures as specified in R1AP [4] that are required to perform testing of the applicable R1 service test cases.

4.2.2.3 Test simulator

For enabling conformance testing, the test simulator has implemented API Producer and/or API Consumer functionality, have HTTP Client and HTTP Server capabilities, and have flexibility to generate, receive, and validate HTTP messages for all the R1 procedures. The test simulator logs all message content during the testing.

As illustrated in figure 4.2.2.1-1, test simulator has all the capabilities required to execute the conformance testing of an rApp. For example, the test simulator has the capabilities to simulate the functionality of A1, O1, and O2 in a cloud environment.

4.3 Conformance testing of SMO/Non-RT RIC framework

4.3.1 General

For conformance testing of SMO/Non-RT RIC framework, SMO/Non-RT RIC framework is the device under test and rApp is the test simulator.

The present document specifies conformance tests for API Consumer and API Producer functionality as specified in R1AP [4].

4.3.2 Test configuration

4.3.2.1 Overview

The test configuration for R1 conformance testing of SMO/Non-RT RIC framework is illustrated in figure 4.3.2.1-1.

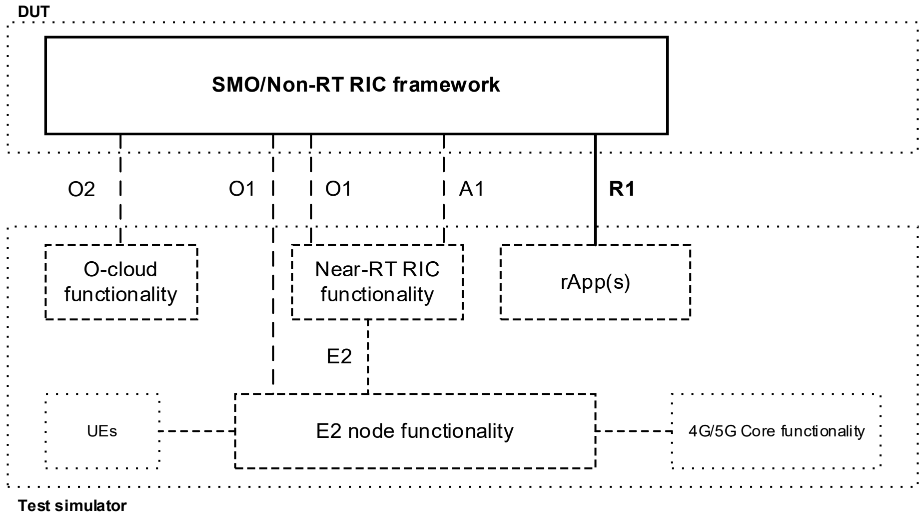

Figure 4.3.2.1-1: Illustration of R1 conformance testing of Non-RT RIC framework.

4.3.2.2 Device under test (SMO/Non-RT RIC Framework)

For enabling conformance testing, the SMO/Non-RT RIC framework has implemented API Producer and/or API Consumer functionality, and the procedures specified in R1AP [4], that are required to perform testing of the applicable test cases.

4.3.2.3 Test simulator

For enabling conformance testing, the test simulator rApp(s) has both API Producer and/or API Consumer and HTTP Server capabilities and have flexibility to generate, receive and validate HTTP messages for all the R1 procedures.

As illustrated in figure 4.3.2.1-1, test simulator has all the capabilities required to execute the conformance testing of an SMO/Non-RT RIC framework. For example, the test simulator has the capabilities to simulate the functionality of A1, O1, and O2.

5 SME Service Registration service test cases

5.1 Conformance test cases for rApp

5.1.1 General

5.1.1.1 Device under test requirements

The rApp that acts as Device Under Test (DUT) in these test scenarios, the requirements on the DUT for these tests are that it can handle the SME service registration service, and the purpose of the test scenarios is to validate that it confirms API Consumer functionality as specified in R1AP [4], clause 6.1.4.

5.1.1.2 Test simulator capabilities

The test simulator has the capabilities as required for a SMO/Non-RT RIC framework. In addition, it has the following capabilities:

- Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.
- Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT's conformance API Producer functionality as specified in R1AP [4], clause 6.1.4.

5.1.2 Register service API as API Consumer test scenario

5.1.2.1 Register service (positive case)

5.1.2.1.1 Test description and applicability

The purpose of this test case is to test the Register service API as specified in R1AP [4], clause 6.1.4.1. The expected outcome is successful validation of the request from the DUT.

5.1.2.1.2 Test entrance criteria

The DUT has functionality to initiate the Register service procedure as defined in R1GAP [1], clause 5.1.3.2.

NOTE: The DUT provides the service API description as defined in R1AP [4], clause A.2.1.3. The service being registered can be a standard or non-standard R1 service.

5.1.2.1.3 Test methodology

5.1.2.1.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

5.1.2.1.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to register a service that it produces.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 6.1.5.2.

b) The HTTP request is a POST operation.

c) The HTTP request message content includes rApp identifier and the service API description conforms to the schema as specified in R1AP [4], clause 6.1.4.1.

Step 4. The test simulator generates the service API identifier and constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 6.1.4.1.1.

5.1.2.1.4 Expected result

The test is considered passed if Step 3 validation has passed.

5.1.2.2 Update registered service (positive case)

5.1.2.2.1 Test description and applicability

The purpose of this test case is to test the Update registered service API as specified in R1AP [4], clause 6.1.4.2. The expected outcome is successful validation of the request from the DUT.

5.1.2.2.2 Test entrance criteria

a) The DUT has functionality to initiate the Update registered service procedure as defined in R1GAP [1], clause 5.1.3.2.

b) A service API registration exists in test simulator and the serviceApiId and the apfId are known to DUT.

c) The schema of the ServiceAPIDescription used for this test are available and used in DUT to formulate the Update registered service request, and in test simulator to validate the request.

NOTE: The DUT provides the ServiceApiDescription as defined in R1AP [4], clause A.2.1.3. The service being updated can be a standard or non-standard R1 service.

5.1.2.2.3 Test methodology

5.1.2.2.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

5.1.2.2.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to update a registered service API identified by the apfId and the serviceApiId.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 6.1.5.3.

b) The HTTP request is a PUT operation.

c) The apfId and the serviceApiId in the URI match the service API being updated.

d) The HTTP request message body contains the ServiceAPIDescription of the service API to be updated and conforms to the schema as specified in R1AP [4], clause 6.1.4.2.

Step 4. The test simulator updates the resource, and a representation of the updated resource shall be returned in the response body the appropriate HTTP response as specified in R1AP [4], clause 6.1.5.3.1.1.

5.1.2.2.4 Expected result

The test is considered passed if Step 3 validation has passed.

5.1.2.3 Deregister service (positive case)

5.1.2.3.1 Test description and applicability

The purpose of this test case is to test the Deregister service API as specified in R1AP [4], clause 6.1.4.3. The expected outcome is successful validation of the request from the DUT.

5.1.2.3.2 Test entrance criteria

1) The DUT has functionality to initiate the Deregister service procedure as defined in R1GAP [1], clause 5.1.3.2.

2) A service API registration exists in test simulator and the serviceApiId and the apfId are known to DUT.

5.1.2.3.3 Test methodology

5.1.2.3.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

5.1.2.3.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to deregister a service API identified by the apfID and the serviceApiId.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 6.1.5.3.

b) The HTTP request is a DELETE operation.

c) The serviceApiId and apfId in the URI match the service being deleted.

d) The message body is empty.

Step 4. The test simulator generated the appropriate HTTP response as specified in R1AP [4], clause 6.1.5.3.1.2.

5.1.2.3.4 Expected result

The test is considered passed if Step 3 validation has passed.

5.1.2.4 Query registered service (positive case)

5.1.2.4.1 Test description and applicability

The purpose of this test case is to test the Query service APIs as specified in R1AP [4], clause 6.1.4.5. The expected outcome is successful validation of the request from the DUT.

5.1.2.4.2 Test entrance criteria

1) The DUT has functionality to initiate Query registered service procedure.

2) A set of service API registrations exist in the test simulator and the apfId is known to DUT.

5.1.2.4.3 Test methodology

5.1.2.4.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

5.1.2.4.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to query service APIs that are registered by the apfId.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 6.1.5.3.

b) The HTTP request is a GET operation.

c) The apfId in the URI match the service APIs being queried.

d) The message body is empty.

Step 4. The test simulator generated the appropriate HTTP response as specified in R1AP [4], clause 6.1.5.2.1.2

5.1.2.4.4 Expected result

The test is considered passed if Step 3 validation has passed.

5.1.5.5 Partially updated registered service (positive case)

5.1.5.5.1 Test description and applicability

The purpose of this test case is to test the partially update registered service API as specified in R1AP [4], clause 6.1.4.4. The expected outcome is successful validation of the request from the DUT.

5.1.5.5.2 Test entrance criteria

1) The DUT has functionality to initiate the Partially update registered service procedure as defined in R1GAP [1], clause 5.1.3.2.

2) A service API registration exists in test simulator and the serviceApiId and the apfId are known to DUT

3) The schema of the ServiceAPIDescriptionPatch used for this test are available and used in DUT to formulate the Partially update registered service request, and in test simulator to validate the request.

NOTE: The DUT provides the ServiceApiDescriptionPatch as defined in R1AP [4]. The service being updated can be a standard or non-standard R1 service.

5.1.5.5.3 Test methodology

5.1.5.5.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

5.1.5.5.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to partially update a registered service API identified by the apfId and the serviceApiId.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 6.1.5.3.

b) The HTTP request is a PATCH operation.

c) The apfId and the serviceApiId in the URI match the service API being updated.

d) The HTTP request message body contains the ServiceAPIDescriptionPatch of the service API to be updated and conforms to the schema as specified in R1AP [4], clause 6.1.4.4.

Step 4. The test simulator updates the resource, and a representation of the partially updated resource shall be returned in the response body the appropriate HTTP response as specified in R1AP [4], clause 6.1.5.3.1.3.

5.1.5.5.4 Expected result

The test is considered passed if Step 3 validation has passed.

5.2 Conformance test cases for SMO/Non-RT RIC framework

5.2.1 General

5.2.1.1 Device under test requirements

The SMO/Non-RT RIC framework that acts as Device Under Test (DUT) in these test scenarios, the requirements on the DUT for these tests are that it can handle the SME service registration service, and the purpose of the test scenarios is to validate that it confirms API Producer functionality as specified in R1AP [4], clause 6.1.4.

5.2.1.2 Test simulator capabilities

The test simulator has the capabilities as specified in section 4.3.2. In addition, it has the following capabilities:

1) Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.

2) Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT's conformance to API Producer functionality as specified in R1AP [4], clause 6.1.4.

5.2.2 Register service API as API Producer test scenario

5.2.2.1 Register service (positive case)

5.2.2.1.1 Test description and applicability

The purpose of this test case is to test the Register service API as specified in R1AP [4], clause 6.1.4.1. The expected outcome is successful validation of the response from the DUT.

5.2.2.1.2 Test entrance criteria

The test simulator has functionality to initiate the Register service procedure as defined in R1GAP [1], clause 5.1.3.2.

NOTE: The test simulator provides the service API description as defined in R1AP [4], clause A.2.1.3. The service being registered can be a standard or non-standard R1 service.

5.2.2.1.3 Test methodology

5.2.2.1.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

5.2.2.1.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to register a service that it produces.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The HTTP request is a POST request.

b) The URI conforms to the format specified in R1AP [4], clause 6.1.5.2.1.1.

c) The HTTP request message content includes the information as specified in R1AP [4], clause 6.1.4.1.

Step 4. The DUT generates the service API identifier and constructs the URI for the created resource.

Step 5. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 6.1.4.1.1.

Step 6. At the test simulator the contents of the received HTTP response is recorded.

Step 7. The test simulator does the following validation:

a) The HTTP response message content includes the information as specified in R1AP [4], clause 6.1.4.1, and the Location header will be present.

b) The URI confirms to the format specified in R1AP [4], clause 6.1.5.2.

5.2.2.1.4 Expected result

The test is considered passed if Step 7 validation has passed.

5.2.2.2 Update registered service (positive case)

5.2.2.2.1 Test description and applicability

The purpose of this test case is to test the Update registered service API as specified in R1AP [4], clause 6.1.4.2. The expected outcome is successful validation of the response from the DUT.

5.2.2.2.2 Test entrance criteria

1) The test simulator has functionality to initiate the Update registered service procedure as defined in R1GAP [1], clause 5.1.3.2.

2) A service API registration exists in DUT and the serviceApiId and the apfId are known to test simulator.

3) The schema of the ServiceAPIDescription used for this test are available and used in test simulator to formulate the Update registered service request, and in the test simulator to validate the request.

NOTE: The test simulator provides the ServiceApiDescription as defined in R1AP [4], clause A.2.1.3. The service being updated can be a standard or non-standard R1 service.

5.2.2.2.3 Test methodology

5.2.2.2.3.1 Initial conditions

The DUT as API Producer is ready and available to receive HTTP requests from the test simulator.

5.2.2.2.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to update a registered service API identified by the apfId and the serviceApiId.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 6.1.5.3.

b) The HTTP request is a PUT operation.

c) The apfId and the serviceApiId in the URI match the service API being updated.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 6.1.4.2.

Step 4. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 6.1.4.2.1.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The HTTP response message content includes the information as specified in R1AP [4], clause 6.1.4.2, and the Location header will be present.

5.2.2.2.4 Expected result

The test is considered passed if Step 6 validation has passed.

5.2.2.3 Deregister service (positive case)

5.2.2.3.1 Test description and applicability

The purpose of this test case is to test the Deregister service API as specified in R1AP [4], clause 6.1.4.3. The expected outcome is successful validation of the response from the DUT.

5.2.2.3.2 Test entrance criteria

1) The test simulator has functionality to initiate the Deregister service procedure as defined in R1GAP [1], clause 5.1.3.2.

2) A service API registration exists in DUT and the serviceApiId and the apfId are known to test simulator.

5.2.2.3.3 Test methodology

5.2.2.3.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP requests from the test simulator.

5.2.2.3.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to deregister a service API identified by the apfID and the serviceApiId.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The HTTP request is a DELETE request.

b) The URI format as specified in R1AP [4]R1AP [4], clause 6.1.5.3.1.2.

c) The HTTP request content includes the information as specified in R1AP [4], clause 6.1.4.3.1.

Step 4. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 6.1.4.3.1.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 6.1.5.3.

b) The HTTP response message content includes the information as specified in R1AP [4], clause 6.1.4.3.1.

c) The message body is empty.

5.2.2.3.4 Expected result

The test is considered passed if Step 6 validation has passed.

5.2.2.4 Query registered service (positive case)

5.2.2.4.1 Test description and applicability

The purpose of this test case is to test the Query service APIs as specified in R1AP [4], clause 6.1.4.5. The expected outcome is successful validation of the response from the DUT.

5.2.2.4.2 Test entrance criteria

1) The test simulator has functionality to initiate Query registered service procedure as defined in R1GAP [1], clause 5.1.3.2.
2) A set of service API registrations exist in the DUT and the apfId is known to test simulator.

5.2.2.4.3 Test methodology

5.2.2.4.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

5.2.2.4.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to query service APIs that are registered by the apfId.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The HTTP request is a GET request.

b) The URI format as specified in R1AP [4]R1AP [4], clause 6.1.5.2.1.2.

c) The HTTP request message content includes the information as specified in R1AP [4], clause 6.1.4.5.1

Step 4. The DUT initiates a HTTP Response.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 6.1.5.2.

b) The HTTP response message content includes the information as specified in R1AP [4], clause 6.1.4.5.1.

5.2.2.4.4 Expected result

The test is considered passed if Step 6 validation has passed.

5.2.5.5 Partially update registered service (positive case)

5.2.5.5.1 Test description and applicability

The purpose of this test case is to test the partially update registered service API as specified in R1AP [4], clause 6.1.4.4. The expected outcome is successful validation of the response from the DUT.

5.2.5.5.2 Test entrance criteria

3) The test simulator has functionality to initiate the Partially update registered service procedure as defined in R1GAP [1], clause 5.1.3.2.

4) A service API registration exists in DUT and the serviceApiId and the apfId are known to test simulator

5) The schema of the ServiceAPIDescriptionPatch used for this test are available and used in test simulator to formulate the Partially update registered service request, and in test simulator to validate the request.

NOTE: The test simulator provides the ServiceApiDescriptionPatch as defined in R1AP [4]. The service being updated can be a standard or non-standard R1 service.

5.2.5.5.3 Test methodology

5.2.5.5.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

5.2.5.5.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to partially update a registered service API identified by the apfId and the serviceApiId.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The HTTP request is a PATCH request.

b) The URI format as specified in R1AP [4], clause 6.1.5.3.1.3.

c) The HTTP request message content includes the information as specified in R1AP [4], clause 6.1.4.4.

Step 4. The DUT initiates a HTTP Response.

Step 5. At the test simulator the contents of the received HTTP request is recorded.

Step 6. The test simulator does the following validation:

a) The HTTP response message content includes the information as specified in R1AP [4], clause 6.1.4.4, and the Location header will be present.

5.2.5.5.4 Expected result

The test is considered passed if Step 6 validation has passed.

6 SME Service Discovery service test cases

6.1 Conformance test cases for rApp

6.1.1 General

6.1.1.1 Device under test requirements

The rApp that acts as Device Under Test (DUT) in these test scenarios, the requirements on the DUT for these tests are that it can handle the SME service discovery service, and the purpose of the test scenarios is to validate that it conforms to the Service discovery API specified in R1AP [4], clause 6.2.4.

6.1.1.2 Test simulator capabilities

The test simulator has the capabilities as required for a SMO/Non-RT RIC framework. In addition, it has the following capabilities:

- Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.
- Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT's conformance API Producer functionality as specified in R1AP [4], clause 6.2.4.

6.1.2 Discover service API as API Consumer test scenario

6.1.2.1 Query service (positive case)

6.1.2.1.1 Test description and applicability

The purpose of this test case is to test the Query service APIs operation as specified in R1AP [4], clause 6.2.4.1. The expected outcome is successful validation of the request from the DUT.

6.1.2.1.2 Test entrance criteria

1) The DUT has functionality to initiate the Discover services procedure.
2) A set of service API registrations exist in the test simulator.

6.1.2.1.3 Test methodology

6.1.2.1.3.1 Initial conditions

1) The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

6.1.2.1.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to query service APIs.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 6.2.5.2.

b) The HTTP request is a GET operation.

c) The message body is empty.

Step 4. The test simulator generated the appropriate HTTP response as specified in R1AP [4], clause 6.2.5.2.1.1.

NOTE: Presence or validation of optional filter parameters is not used to determine validation on this test.

6.1.2.1.4 Expected result

The test is considered passed if Step 3 validation has passed.

6.2 Conformance test cases for SMO/Non-RT RIC framework

6.2.1 General

6.2.1.1 Device under test requirements

The SMO/Non-RT RIC framework that acts as Device Under Test (DUT) in these test scenarios, the requirements on the DUT for these tests are that it can handle the SME service discovery service, and the purpose of the test scenarios is to validate that it conforms to the Service discovery API specified in R1AP [4], clause 6.2.4.

6.2.1.2 Test simulator capabilities

The test simulator has the capabilities as specified in section 4.3.2. In addition, it has the following capabilities:

1) Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.

2) Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT's conformance to API Producer functionality as specified in R1AP [1], clause 6.2.4.

6.2.2 Service discovery API as API Producer test scenario

6.2.2.1 Query service (positive case)

6.2.2.1.1 Test description and applicability

The purpose of this test case is to test the Query service APIs as specified in R1AP [1] clause 6.2.4.1. The expected outcome is successful validation of the response from the DUT.

6.2.2.1.2 Test entrance criteria

1) The DUT has functionality to initiate the Discover services procedure.

2) A set of service API registrations exist in the test simulator.

6.2.2.1.3 Test methodology

6.2.2.1.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

6.2.2.1.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to query service APIs that are registered by the apfId.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The HTTP request is a GET request.

b) The URI conforms to the format as specified in R1AP[4], clause 6.2.5.2.

c) The HTTP request message content includes the information as specified in R1AP [4], clause 6.2.4.1.

Step 4. The DUT initiates a HTTP Response.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP[4], clause 6.2.5.2.

b) The HTTP response message content includes the information as specified in R1AP[4], clause.6.2.4.1.

6.2.2.1.4 Expected result

The test is considered passed if Step 6 validation has passed.

7 SME Service Subscription service test cases

7.1 Conformance test cases for rApp

7.1.1 General

7.1.1.1 Device under test requirements

The rApp that acts as DUT in these test scenarios, the requirements on the DUT for these tests are that it can handle the SME service subscription service, and the purpose of the test scenarios is to validate that it conforms to the API Consumer functionality as specified in R1AP [4], clause 6.3.4.

7.1.1.2 Test simulator capabilities

The test simulator has the capabilities as required for a SMO/Non-RT RIC framework. In addition, it has the following capabilities:

- Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.
- Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT's conformance to the API Consumer functionality as specified in R1AP[4], clause 6.3.4.

7.1.2 Service events subscription API as API Consumer test scenario

7.1.2.1 Subscribe service events (positive case)

7.1.2.1.1 Test description and applicability

The purpose of this test case is to test the Events subscription service API as specified in R1AP[4], clause 6.3.4.1. The expected outcome is successful validation of the request from the DUT.

7.1.2.1.2 Test entrance criteria

The DUT has functionality to initiate the Subscribe service availability procedure as defined in R1GAP [1], clause 5.1.4.

NOTE: The DUT provides the events subscription structure as defined in R1AP[4], clause 6.3.3. The service event being subscribed to can be a standard or non-standard R1 service event.

7.1.2.1.3 Test methodology

7.1.2.1.3.1 Initial conditions

1) The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.
2) The subscriberId is known to DUT.

7.1.2.1.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to subscribe to service event notifications.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP[4], clause 6.3.5.2.
b) The HTTP request is a POST operation.
c) The HTTP request message content includes the information as specified in R1AP[4], clause 6.3.4.1.1.

Step 4. The test simulator generates the subscription identifier and constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP[4], clause 6.3.4.1.1.

7 .1.2.1.4 Expected result

The test is considered passed if Step 3 validation has passed.

7.1.2.2 Unsubscribe service events (positive case)

7.1.2.2.1 Test description and applicability

The purpose of this test case is to test the unsubscribe service API as specified in R1AP[4], clause 6.3.4.2. The expected outcome is successful validation of the request from the DUT.

7.1.2.2.2 Test entrance criteria

1) The DUT has functionality to initiate the Unsubscribe service availability procedure as defined in R1GAP [1], clause 5.1.4.
2) A subscription exists and the subscriberId and subscriptionId are known to the DUT.

7.1.2.2.3 Test methodology

7.1.2.2.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

7.1.2.2.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to unsubscribe from service event notifications identified by the subscriberId and subscriptionId.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP[4], clause 6.3.5.3.

b) The HTTP request is a DELETE operation.

c) The subscriberId and subscriptionId in the URI match the service events being deleted.

d) The message body is empty.

Step 4. The test simulator generated the appropriate HTTP response as specified in R1AP[4], clause 6.3.5.3.1.1.

7.1.2.2.4 Expected result

The test is considered passed if Step 3 validation has passed.

7.2 Conformance test cases for SMO/Non-RT RIC framework

7.2.1 General

7.2.1.1 Device under test requirements

The SMO/Non-RT RIC framework that acts as DUT in these test scenarios, the requirements on the DUT for these tests are that it can handle the SME service events subscription service, and the purpose of the test scenarios is to validate that it conforms to the API Producer functionality as specified in R1AP [4], clause 6.3.4.

7.2.1.2 Test simulator capabilities

The test simulator has the capabilities as specified in section 4.3.2. In addition, it has the following capabilities:

1) Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.

2) Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT’s conformance to the API Producer functionality as specified in R1AP[4], clause 6.3.4.

7.2.2 Service events subscription API as API Producer test scenario

7.2.2.1 Service events subscription (positive case)

7.2.2.1.1 Test description and applicability

The purpose of this test case is to test the Events subscription service API as specified in R1AP[4], clause 6.3.4.1. The expected outcome is successful validation of the response from the DUT.

7.2.2.1.2 Test entrance criteria

The test simulator has functionality to initiate the Subscribe service availability procedure as defined in R1GAP [1], clause 5.1.4.

NOTE: The test simulator provides the events subscription structure as defined in R1AP[4], clause 6.3.3. The service being subscribed to can be a standard or non-standard R1 service.

7.2.2.1.3 Test methodology

7.2.2.1.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

7.2.2.1.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to subscribe to service event notifications.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The HTTP request is a POST request.

b) The URI conforms to the format specified in R1AP[4], clause 6.3.5.2.1.1.

c) The HTTP request message content includes the information as specified in R1AP[4], clause 6.3.4.1.1.

Step 4. The DUT generates the subscription identifier and constructs the URI for the created resource.

Step 5. The DUT sends the appropriate HTTP response as specified in R1AP[4], clause 6.3.4.1.1.

Step 6. At the test simulator the contents of the received HTTP response is recorded.

Step 7. The test simulator does the following validation:

a) The HTTP response message content includes the information as specified in R1AP[4], clause 6.3.4.1.1, and the Location header will be present.

b) The URI conforms to the format specified in R1AP[4], clause 6.3.5.2.

7.2.2.1.4 Expected result

The test is considered passed if Step 7 validation has passed.

7.2.2.2 Unsubscribe service events (positive case)

7.2.2.2.1 Test description and applicability

The purpose of this test case is to test the Unsubscribe service API as specified in R1AP[4], clause 6.3.4.2. The expected outcome is successful validation of the response from the DUT.

7.2.2.2.2 Test entrance criteria

1) The test simulator has functionality to initiate the Unsubscribe service availability procedure as defined in R1GAP [1], clause 5.1.4.

2) A subscription exists and the subscriberId and subscriptionId are known to the test simulator.

7.2.2.2.3 Test methodology

7.2.2.2.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP requests from the DUT.

7.2.2.2.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to unsubscribe from service event notifications identified by the subscriberId and subscriptionId.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The HTTP request is a DELETE request.

b) The URI format as specified in R1AP [4], clause 6.3.5.3.1.1.

c) The subscriberId and subscriptionId in the URI match the service events being deleted.

d) The message body is empty.

Step 4. The DUT sends the appropriate HTTP response as specified in R1AP[4] , clause 6.3.4.2.1.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The return code is "204 No content".

b) The response message body is empty.

7.2.2.2.4 Expected result

The test is considered passed if Step 6 validation has passed.

7.2.2.3 Notification service (positive case)

7.2.2.3.1 Test description and applicability

The purpose of this test case is to test the notification service API as specified in R1AP[4], clause 6.3.4.3. The expected outcome is successful validation of the event notification from the DUT.

7.2.2.3.2 Test entrance criteria

1) The DUT has functionality to initiate the Notify service availability changes procedure as defined in R1GAP [1], clause 5.1.4.

2) A subscription exists in the DUT.

7.2.2.3.3 Test methodology

7.2.2.3.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP requests from the DUT.

7.2.2.3.3.2 Procedure

Step 1. The DUT as an API producer initiates the sending of a HTTP POST request.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The HTTP request is a POST operation.

b) The callback URI matches the URI provided in the subscription.

7.2.2.3.4 Expected result

The test is considered passed if Step 3 validation has passed.

8 DME Data registration service test cases

8.1 Conformance test cases for rApp

8.1.1 General

8.1.1.1 Device under test requirements

The rApp that acts as DUT in the test scenarios, the requirements on the DUT for these tests are that it can handle the Data registration service, and the purpose of the test scenarios is to validate that it conforms to the API Consumer functionality as specified in R1AP [4], clause 7.1.4.

8.1.1.2 Test simulator capabilities

The test simulator has the capabilities as required for a SMO/Non-RT RIC framework. In addition, it has the following capabilities:

- Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.
- Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT's conformance to the API Consumer functionality as specified in R1AP [4], clause 7.1.4.

8.1.2 Data registration API as API consumer test scenario

8.1.2.1 Register DME type (positive case)

8.1.2.1.1 Test description and applicability

The purpose of this test case is to test the Register DME type in Data registration API as specified in R1AP [4], clause 7.1.4.1. The expected outcome is successful validation of the request from the DUT.

8.1.2.1.2 Test entrance criteria

The DUT has functionality to initiate the Register DME type procedure as defined in R1GAP [1], clause 5.2.2.

NOTE: The DUT provides the service API description as defined in R1AP [4], clause A.2.1.3. The service being registered can be a standard or non-standard R1 service.

8.1.2.1.3 Test methodology

8.1.2.1.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

8.1.2.1.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to register a DmeTypeRelatedCapabilities.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 7.1.5.2.2.

b) The HTTP request is a POST operation.

c) The HTTP request message content includes DmeTypeRelatedCapabilities as specified in R1AP [4], clause 7.1.4.

Step 4. The test simulator generates the registrationId and constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 7.1.5.2.3.1.

8.1.2.1.4 Expected result

The test is considered passed if Step 3 validation has passed.

8.1.2.2 Deregister DME type (positive case)

8.1.2.2.1 Test description and applicability

The purpose of this test case is to test the Deregister DME type in Data registration API as specified in R1AP [4], clause 7.1.4.2. The expected outcome is successful validation of the request from the DUT.

8.1.2.2.2 Test entrance criteria

1) The DUT has functionality to initiate the Deregister DME type procedure as defined in R1GAP [1], clause 5.2.2.2.

2) A DME type exists in test simulator and the registrationId is known to the DUT.

8.1.2.2.3 Test methodology

8.1.2.2.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

8.1.2.2.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to Deregister DME type.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.1.5.3.2.

b) The HTTP request is a DELETE operation.

c) The registrationId in the URI match the DME type being deregistered.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 7.1.5.3.3.

Step 4. The test simulator constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 7.1.5.3.3

8.1.2.2.4 Expected result

The test is considered passed if Step 3 validation has passed.

8.1.2.3 Query DME type (positive case)

8.1.2.3.1 Test description and applicability

The purpose of this test case is to test the Query DME type in Data registration API as specified in R1AP [4], clause 7.1.4.4. The expected outcome is successful validation of the request from the DUT.

8.1.2.3.2 Test entrance criteria

1) The DUT has functionality to initiate the query DME type registration procedure as defined in R1GAP [1], clause 5.2.2.2.
2) A DME type exists in test simulator and the registrationId is known to the DUT.

8.1.2.3.3 Test methodology

8.1.2.3.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

8.1.2.3.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to query a DME type.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.1.5.3.2.
b) The HTTP request is a GET operation.
c) The registrationId in the URI match the DME type being queried.
d) The HTTP request message content includes the information as specified in R1AP [4], clause 7.1.5.3.3.

Step 4. The test simulator constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 7.1.5.3.3.

8.1.2.3.4 Expected result

The test is considered passed if Step 3 validation has passed.

8.1.2.4 Update DME type (positive case)

8.1.2.4.1 Test description and applicability

The purpose of this test case is to test the Update DME type in Data registration API as specified in R1AP [4], clause 7.1.4.3. The expected outcome is successful validation of the request from the DUT.

8.1.2.4.2 Test entrance criteria

1) The DUT has functionality to initiate the Update DME type registration procedure as defined in R1GAP [1], clause 5.2.2.2.
2) A DME type exists in test simulator and the registrationId and DmeTypeRelatedCapabilities are known to the DUT.

8.1.2.4.3 Test methodology

8.1.2.4.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

8.1.2.4.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to update a DME type identified by the registrationId and DmeTypeRelatedCapabilities.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 7.1.5.3.2.

b) The HTTP request is a PUT operation.

c) The registrationId and DmeTypeRelatedCapabilities in the URI match the DME type being updated.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 7.1.5.3.3.

Step 4. The test simulator updates the resource, and a representation of the updated resource shall be returned in the response body the appropriate HTTP response as specified in R1AP [4], clause 7.1.5.3.3.

8.1.2.4.4 Expected result

The test is considered passed if Step 3 validation has passed.

8.1.2.5 Query data subscription information (positive case)

8.1.2.5.1 Test description and applicability

The purpose of this test case is to test the Query data jobs in the Data registration API as specified in R1AP [4], clause 7.1.4.5. The expected outcome is successful validation of the request from the DUT.

8.1.2.5.2 Test entrance criteria

1) The DUT has functionality to initiate the Query data subscription information procedure as defined in R1GAP [1], clause 5.2.2.
2) A of set of data jobs exist in the test simulator and the registrationId is known to the DUT

8.1.2.5.3 Test methodology

8.1.2.5.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

8.1.2.5.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to query a set of data jobs.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.1.5.4.2.

b) The HTTP request is a GET operation.

c) The registrationId in the URI match the DME type being queried.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 7.1.4.5.1.

Step 4. The test simulator constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 7.1.4.5.1

8.1.2.5.4 Expected result

The test is considered passed if Step 3 validation has passed.

8.2 Conformance test cases for SMO/Non-RT RIC framework

8.2.1 General

8.2.1.1 Device under test requirements

The SMO/Non-RT RIC framework that acts as DUT in these test scenarios, the requirements on the DUT for these tests are that it can handle the Data registration service, and the purpose of the test scenarios is to validate that it conforms to the API Producer functionality as specified in R1AP [4], clause 7.1.4.

8.2.1.2 Test simulator capabilities

The test simulator has the capabilities as required for a rApp. In addition, it has the following capabilities:

- Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.
- Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT's conformance to the API Producer functionality as specified in R1AP [4], clause 7.1.4.

8.2.2 Data registration API as API Producer test scenario

8.2.2.1 Register DME type (positive case)

8.2.2.1.1 Test description and applicability

The purpose of this test case is to test the Register DME type in Data registration API as specified in R1AP [4], clause 7.1.4.1. The expected outcome is successful validation of the request from the DUT.

8.2.2.1.2 Test entrance criteria

The test simulator has functionality to initiate the Register DME type procedure as defined in R1GAP [1], clause 5.2.2.

8.2.2.1.3 Test methodology

8.2.2.1.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

8.2.2.1.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to register a DmeTypeRelatedCapabilities.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The HTTP request is a POST request.

b) The URI conforms to the format specified in R1AP [4], clause 7.1.5.2.2.

c) The HTTP request message content includes the information as specified in R1AP [4], clause 7.1.5.2.3.1.

Step 4. The DUT generates the data job and constructs the URI for the created resource.

Step 5. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 7.1.5.2.3.1.

Step 6. At the test simulator the contents of the received HTTP response is recorded.

Step 7. The test simulator does the following validation:

a) The HTTP response message content includes the information as specified in R1AP [4], clause 7.1.5.2.3.1, and the Location header will be present.

b) The URI conforms to the format specified in R1AP [4], clause 7.1.5.2.2.

8.2.2.1.4 Expected result

The test is considered passed if Step 7 validation has passed.

8.2.2.2 Deregister DME type (positive case)

8.2.2.2.1 Test description and applicability

The purpose of this test case is to test the Deregister DME type in Data registration API as specified in R1AP [4], clause 7.1.4.2. The expected outcome is successful validation of the request from the DUT.

8.2.2.2.2 Test entrance criteria

1) The test simulator has functionality to initiate the Deregister DME type procedure as defined in R1GAP [1] clause 5.2.2.2.

2) A DME type exists in test simulator and the registrationId is known to the DUT.

8.2.2.2.3 Test methodology

8.2.2.2.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

8.2.2.2.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to Deregister DME type.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.1.5.3.2.

b) The HTTP request is a DELETE operation.

c) The registrationId in the URI match the DME type being deregistered.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 7.1.5.3.3.

Step 4. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 7.1.5.3.3.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 7.1.5.3.2.

b) The HTTP response message content includes the information as specified in R1AP [4], clause 7.1.5.3.3.

8.2.2.2.4 Expected result

The test is considered passed if Step 6 validation has passed.

8.2.2.3 Query DME type (positive case)

8.2.2.3.1 Test description and applicability

The purpose of this test case is to test the Query DME type in Data registration API as specified in R1AP [4], clause 7.1.4.4. The expected outcome is successful validation of the request from the DUT.

8.2.2.3.2 Test entrance criteria

1) The test simulator has functionality to initiate the query DME type registration procedure as defined in R1GAP [1], clause 5.2.2.2.

2) A DME type exists in test simulator and the registrationId is known to the DUT.

8.2.2.3.3 Test methodology

8.2.2.3.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

8.2.2.3.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to query a DME type.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.1.5.3.2.

b) The HTTP request is a GET operation.

c) The registrationId in the URI match the data job status being queried.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 7.1.5.3.3.

Step 4. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 7.1.5.3.3.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.1.5.3.2.

b) The HTTP response message content includes the information as specified in R1AP [4], clause 7.1.5.3.3.

8.2.2.3.4 Expected result

The test is considered passed if Step 6 validation has passed.

8.2.2.4 Update DME type (positive case)

8.2.2.4.1 Test description and applicability

The purpose of this test case is to test the Update DME type in Data registration API as specified in R1AP [4], clause 7.1.4.3. The expected outcome is successful validation of the request from the DUT.

8.2.2.4.2 Test entrance criteria

1) The test simulator has functionality to initiate the Update DME type registration procedure as defined in R1GAP [1], clause 5.2.2.2

2) A DME type exists in test simulator and the registrationId and DmeTypeRelatedCapabilities are known to the DUT

8.2.2.4.3 Test methodology

8.2.2.4.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

8.2.2.4.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to update a DME type identified by the registrationId and DmeTypeRelatedCapabilities.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The HTTP request is a PUT request.

b) The URI conforms to the format specified in R1AP [4], clause 7.1.5.3.2.

c) The HTTP request message content includes the information as specified in R1AP [4], clause 7.1.5.3.3.

Step 4. The DUT generates the data job and constructs the URI for the created resource.

Step 5. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 7.1.5.3.3.

Step 6. At the test simulator the contents of the received HTTP response is recorded.

Step 7. The test simulator does the following validation:

a) The HTTP response message content includes the information as specified in R1AP [4], clause 7.1.5.3.3.

b) The URI conforms to the format specified in R1AP [4], clause 7.1.5.3.2.

8.2.2.4.4 Expected result

The test is considered passed if Step 7 validation has passed.

8.2.2.5 Query data subscription information (positive case)

8.2.2.5.1 Test description and applicability

The purpose of this test case is to test the Query data jobs in the Data registration API as specified in R1AP [4], clause 7.1.4.5. The expected outcome is successful validation of the request from the DUT.

8.2.2.5.2 Test entrance criteria

1) The test simulator has functionality to initiate the Query data subscription information procedure as defined in R1GAP [1], clause 5.2.2.
2) A of set of data jobs exist in the DUT and the registrationId is known to the test simulator

8.2.2.5.3 Test methodology

8.2.2.5.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

8.2.2.5.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to query a set of data jobs.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.1.5.4.2.
b) The HTTP request is a GET operation.
c) The registrationId in the URI match the DME Type being queried.
d) The HTTP request message content includes the information as specified in R1AP [4], clause 7.1.4.5.1.

Step 4. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 7.1.4.5.1

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.1.5.4.2
b) The HTTP response message content includes the information as specified in R1AP [4], clause 7.1.4.5.1.

8.2.2.5.4 Expected result

The test is considered passed if Step 6 validation has passed.

9 DME Data request service test cases

9.1 Conformance test cases for rApp

9.1.1 General

9.1.1.1 Device under test requirements

The rApp that acts as DUT in these test scenarios, the requirements on the DUT for these tests are that it can handle the DME Data request service, and the purpose of the test scenarios is to validate that it conforms to the API Consumer functionality as specified in R1AP [4], clause 7.3.4.

9.1.1.2 Test simulator capabilities

The test simulator has the capabilities as required for a SMO/Non-RT RIC framework. In addition, it has the following capabilities:

- Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.
- Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT's conformance to the API Consumer functionality as specified in R1AP [4], clause 7.3.4.

9.1.2 Data access API as API Consumer test scenario

9.1.2.1 Create data job (positive case)

9.1.2.1.1 Test description and applicability

The purpose of this test case is to test the create data job API as specified in R1AP [4], clause 7.3.4.1. The expected outcome is successful validation of the request from the DUT.

9.1.2.1.2 Test entrance criteria

1) The DUT has functionality to initiate the request data procedure as defined in R1GAP [1], clause 5.2.4.
2) The DataJobInfo is known to the DUT.

9.1.2.1.3 Test methodology

9.1.2.1.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

9.1.2.1.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to create a data job.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.3.4.1.

b) The HTTP request is a POST operation.

c) The HTTP request message content includes the information as specified in R1AP [4], clause 7.3.4.1.

Step 4. The test simulator constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [1], clause 7.3.4.1.1.

9.1.2.1.4 Expected result

The test is considered passed if Step 3 validation has passed.

9.1.2.2 Cancel data job (positive case)

9.1.2.2.1 Test description and applicability

The purpose of this test case is to test the Cancel data job API as specified in R1AP [4], clause 7.3.4.2. The expected outcome is successful validation of the request from the DUT.

9.1.2.2.2 Test entrance criteria

1) The DUT has functionality to initiate the cancel data request procedure as defined in R1GAP [1], clause 5.2.4.
2) A data job exists in test simulator and the dataJobId is known to the DUT.

9.1.2.2.3 Test methodology

9.1.2.2.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

9.1.2.2.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to cancel a data job.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.3.5.3.

b) The HTTP request is a DELETE operation.

c) The dataJobID in the URI match the data job being deleted.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 7.3.4.2.

Step 4. The test simulator constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [1], clause 7.3.4.2.1.

9.1.2.2.4 Expected result

The test is considered passed if Step 3 validation has passed.

9.1.2.3 Query data job status (positive case)

9.1.2.3.1 Test description and applicability

The purpose of this test case is to test the query data job status API as specified in R1AP [4], clause 7.3.4.6 The expected outcome is successful validation of the request from the DUT.

9.1.2.3.2 Test entrance criteria

1) The DUT has functionality to initiate the query data request status procedure as defined in R1GAP [1], clause 5.2.4.
2) A data job exists in test simulator and the dataJobId is known to the DUT.

9.1.2.3.3 Test methodology

9.1.2.3.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

9.1.2.3.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to query a data job status.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.3.4.6.1.
b) The HTTP request is a GET operation.
c) The dataJobID in the URI match the data job being queried.
d) The HTTP request message content includes the information as specified in R1AP [4], clause 7.3.4.6.1.

Step 4. The test simulator constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 7.3.4.6.1.

9.1.2.3.4 Expected result

The test is considered passed if Step 3 validation has passed.

9.2 Conformance test cases for SMO/Non-RT RIC framework

9.2.1 General

9.2.1.1 Device under test requirements

The SMO/Non-RT RIC framework that acts as DUT in these test scenarios, the requirements on the DUT for these tests are that it can handle the DME Data request service, and the purpose of the test scenarios is to validate that it conforms to the API Producer functionality as specified in R1AP [4], clause 7.3.4.

9.2.1.2 Test simulator capabilities

The test simulator has the capabilities as required for a rApp. In addition, it has the following capabilities:

- Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.
- Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT’s conformance to the API Producer functionality as specified in R1AP [4], clause 7.3.4.

9.2.2 Data access API as API Producer test scenario

9.2.2.1 Create data job (positive case)

9.2.2.1.1 Test description and applicability

The purpose of this test case is to test the create data job API as specified in R1AP [4], clause 7.3.4.1. The expected outcome is successful validation of the request from the DUT.

9.2.2.1.2 Test entrance criteria

1) The test simulator has functionality to initiate the request data procedure as defined in R1GAP [1], clause 5.2.4.
2) The DataJobInfo is known to the DUT

9.2.2.1.3 Test methodology

9.2.2.1.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

9.2.2.1.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to create a data job.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The HTTP request is a POST request.

b) The URI conforms to the format specified in R1AP [4], clause 7.3.5.2.3.1.

c) The HTTP request message content includes the information as specified in R1AP [4], clause 7.3.4.1.

Step 4. The DUT generates the data job and constructs the URI for the created resource.

Step 5. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 7.3.4.1.

Step 6. At the test simulator the contents of the received HTTP response is recorded.

Step 7. The test simulator does the following validation:

a) The HTTP response message content includes the information as specified in R1AP [4], clause 7.3.4.1, and the Location header will be present.

b) The URI conforms to the format specified in R1AP [4], clause 7.3.5.3.

9.2.2.1.4 Expected result

The test is considered passed if Step 7 validation has passed.

9.2.2.2 Cancel data job (positive case)

9.2.2.2.1 Test description and applicability

The purpose of this test case is to test the Cancel data job API as specified in R1AP [4], clause 7.3.4.2. The expected outcome is successful validation of the request from the DUT.

9.2.2.2.2 Test entrance criteria

1) The test simulator has functionality to initiate the cancel data request procedure as defined in R1GAP [1], clause 5.2.4.

2) A data job exists in DUT and the dataJobId is known to the test simulator.

9.2.2.2.3 Test methodology

9.2.2.2.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

9.2.2.2.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to cancel a data job.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.3.5.3.3.1.

b) The HTTP request is a DELETE operation.

c) The dataJobID in the URI match the data job being deleted.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 7.3.4.2.

Step 4. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 7.3.4.2.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 7.3.5.3.

b) The HTTP response message content includes the information as specified in R1AP [4], clause 7.3.4.2.

9.2.2.2.4 Expected result

The test is considered passed if Step 6 validation has passed.

9.2.2.3 Query data job status (positive case)

9.2.2.3.1 Test description and applicability

The purpose of this test case is to test the query data job status API as specified in R1AP [4], clause 7.3.4.6 The expected outcome is successful validation of the request from the DUT.

9.2.2.3.2 Test entrance criteria

1) The test simulator has functionality to initiate the query data request status procedure as defined in R1GAP [1], clause 5.2.4.

2) A data job exists in the DUT and the dataJobId is known to the test simulator.

9.2.2.3.3 Test methodology

9.2.2.3.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

9.2.2.3.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to query a data job status.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.3.5.4.

b) The HTTP request is a GET operation.

c) The dataJobID in the URI match the data job status being queried.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 7.3.4.6.1.

Step 4. The DUT constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 7.3.4.6.1.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.3.5.4.

b) The HTTP response message content includes the information as specified in R1AP [4], clause 7.3.4.6.1.

9.2.2.3.4 Expected result

The test is considered passed if Step 6 validation has passed.

10 DME Data subscription service test cases

10.1 Conformance test cases for rApp

10.1.1 General

10.1.1.1 Device under test requirements

The rApp that acts as DUT in these test scenarios, the requirements on the DUT for these tests are that it can handle the DME Data subscription service, and the purpose of the test scenarios is to validate that it conforms to the API Consumer functionality as specified in R1AP [4], clause 7.3.4.

10.1.1.2 Test simulator capabilities

The test simulator has the capabilities as required for a SMO/Non-RT RIC framework. In addition, it has the following capabilities:

- Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.

- Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT's conformance to the API Consumer functionality as specified in R1AP [4], clause 7.3.4.

10.1.2 Data access API as API Consumer test scenario

10.1.2.1 Create data job (positive case)

10.1.2.1.1 Test description and applicability

The purpose of this test case is to test the create data job API as specified in R1AP [4], clause 7.3.4.1. The expected outcome is successful validation of the request from the DUT.

10.1.2.1.2 Test entrance criteria

1) The DUT has functionality to initiate the subscribe data procedure as defined in R1GAP [1], clause 5.2.5.
2) The DataJobInfo is known to the DUT

10.1.2.1.3 Test methodology

10.1.2.1.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

10.1.2.1.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to create a data job.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.3.4.1.
b) The HTTP request is a POST operation.
c) The HTTP request message content includes the information as specified in R1AP [4], clause 7.3.4.1.

Step 4. The test simulator constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 7.3.4.1.1.

10.1.2.1.4 Expected result

The test is considered passed if Step 3 validation has passed.

10.1.2.2 Cancel data job (positive case)

10.1.2.2.1 Test description and applicability

The purpose of this test case is to test the Cancel data job API as specified in R1AP [4], clause 7.3.4.2. The expected outcome is successful validation of the request from the DUT.

10.1.2.2.2 Test entrance criteria

1) The DUT has functionality to initiate the unsubscribe data procedure as defined in R1GAP [1], clause 5.2.5.

2) A data job exists in test simulator and the dataJobId is known to the DUT.

10.1.2.2.3 Test methodology

10.1.2.2.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

10.1.2.2.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to cancel a data job.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.3.5.3.

b) The HTTP request is a DELETE operation.

c) The dataJobID in the URI match the data job being deleted.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 7.3.4.2.

Step 4. The test simulator constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 7.3.4.2.1.

10.1.2.2.4 Expected result

The test is considered passed if Step 3 validation has passed.

10.1.2.3 Query data job (positive case)

10.1.2.3.1 Test description and applicability

The purpose of this test case is to test the query data job API as specified in R1AP [4], clause 7.3.4.5 The expected outcome is successful validation of the request from the DUT.

10.1.2.3.2 Test entrance criteria

1) The DUT has functionality to initiate the query data subscription procedure as defined in R1GAP [1], clause 5.2.5.

2) A data job exists in test simulator and the dataJobId is known to the DUT.

10.1.2.3.3 Test methodology

10.1.2.3.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

10.1.2.3.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to query a data job.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.3.4.5.1.

b) The HTTP request is a GET operation.

c) The dataJobID in the URI match the data job being queried.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 7.3.4.5.1.

Step 4. The test simulator constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 7.3.4.6.1.

10.1.2.3.4 Expected result

The test is considered passed if Step 3 validation has passed.

10.1.2.4 Query data job status (positive case)

10.1.2.4.1 Test description and applicability

The purpose of this test case is to test the query data job status API as specified in R1AP [4], clause 7.3.4.6 The expected outcome is successful validation of the request from the DUT.

10.1.2.4.2 Test entrance criteria

1) The DUT has functionality to initiate the query data subscription status procedure as defined in R1GAP [1], clause 5.2.5.

2) A data job exists in test simulator and the dataJobId is known to the DUT.

10.1.2.4.3 Test methodology

10.1.2.4.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

10.1.2.4.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to query a data job status.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.3.4.6.1.

b) The HTTP request is a GET operation.

c) The dataJobID in the URI match the data job being queried.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 7.3.4.6.1.

Step 4. The test simulator constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 7.3.4.6.1.

10.1.2.4.4 Expected result

The test is considered passed if Step 3 validation has passed.

10.1.2.5 Query data job identifiers (positive case)

10.1.2.5.1 Test description and applicability

The purpose of this test case is to test the query data job identifiers API as specified in R1AP [4], clause 7.3.4.7 The expected outcome is successful validation of the request from the DUT.

10.1.2.5.2 Test entrance criteria

1) The DUT has functionality to initiate the query data subscription status procedure as defined in R1GAP [1], clause 5.2.5.
2) A set of data jobs exists in test simulator.

10.1.2.5.3 Test methodology

10.1.2.5.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

10.1.2.5.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to query data job identifiers.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a)The URI conforms to the format specified in R1AP [4], clause 7.3.4.7.1.

b)The HTTP request is a GET operation.

c)The HTTP request message content includes the information as specified in R1AP [4], clause 7.3.4.7.1.

Step 4. The test simulator constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 7.3.4.7.1.

10.1.2.5.4 Expected result

The test is considered passed if Step 3 validation has passed.

10.1.2.6 Update data job (positive case)

10.1.2.6.1 Test description and applicability

The purpose of this test case is to test the Update data job API as specified in R1AP [4], clause 7.3.4.5. The expected outcome is successful validation of the request from the DUT.

10.1.2.6.2 Test entrance criteria

1) The DUT has functionality to initiate the Update data job procedure as defined in R1GAP [1], clause 5.2.5.
2) A data job exists in test simulator and the dataJobId and DataJobInfo are known to the DUT.

10.1.2.6.3 Test methodology

10.1.2.6.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

10.1.2.6.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to update a data job identified by the dataJobId and DataJobInfo.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 7.3.4.4.

b) The HTTP request is a PUT operation.

c) The dataJobId and DataJobInfo in the URI match the data job being updated

d) The HTTP request message content includes the information as specified in R1AP [4], clause 7.3.4.4.1.

Step 4. The test simulator updates the resource, and a representation of the updated resource shall be returned in the response body the appropriate HTTP response as specified in R1AP [4], clause 7.3.4.4.1.

10.1.2.6.4 Expected result

The test is considered passed if Step 3 validation has passed.

10.2 Conformance test cases for SMO/Non-RT RIC framework

10.2.1 General

10.2.1.1 Device under test requirements

The SMO/Non-RT RIC framework that acts as DUT in these test scenarios, the requirements on the DUT for these tests are that it can handle the DME Data subscription service, and the purpose of the test scenarios is to validate that it conforms to the API Producer functionality as specified in R1AP [4], clause 7.3.4.

10.2.1.2 Test simulator capabilities

The test simulator has the capabilities as required for a rApp. In addition, it has the following capabilities:

- Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.
- Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT's conformance to the API Producer functionality as specified in R1AP [4], clause 7.3.4.

10.2.2 Data access API as API Producer test scenario

10.2.2.1 Create data job (positive case)

10.2.2.1.1 Test description and applicability

The purpose of this test case is to test the create data job API as specified in R1AP [4], clause 7.3.4.1. The expected outcome is successful validation of the request from the DUT.

10.2.2.1.2 Test entrance criteria

1) The test simulator has functionality to initiate the subscribe data procedure as defined in R1GAP [1], clause 5.2.5.
2) The DataJobInfo is known to the DUT

10.2.2.1.3 Test methodology

10.2.2.1.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

10.2.2.1.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to create a data job.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The HTTP request is a POST request.

b) The URI conforms to the format specified in R1AP [4], clause 7.3.5.2.3.1.

c) The HTTP request message content includes the information as specified in R1AP [4], clause 7.3.4.1.

Step 4. The DUT generates the data job and constructs the URI for the created resource.

Step 5. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 7.3.4.1.

Step 6. At the test simulator the contents of the received HTTP response is recorded.

Step 7. The test simulator does the following validation:

a. The HTTP response message content includes the information as specified in R1AP [4], clause 7.3.4.1, and the Location header will be present.
b. The URI conforms to the format specified in R1AP [4], clause 7.3.5.3.

10.2.2.1.4 Expected result

The test is considered passed if Step 7 validation has passed.

10.2.2.2 Cancel data job (positive case)

10.2.2.2.1 Test description and applicability

The purpose of this test case is to test the Cancel data job API as specified in R1AP [4], clause 7.3.4.2. The expected outcome is successful validation of the request from the DUT.

10.2.2.2.2 Test entrance criteria

1) The test simulator has functionality to initiate the unsubscribe data procedure as defined in R1GAP [1], clause 5.2.5.
2) A data job exists in DUT and the dataJobId is known to the test simulator.

10.2.2.2.3 Test methodology

10.2.2.2.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

10.2.2.2.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to cancel a data job.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.3.5.3.3.1.

b) The HTTP request is a DELETE operation.

c) The dataJobID in the URI match the data job being deleted.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 7.3.4.2.

Step 4. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 7.3.4.2.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 7.3.5.3.

b) The HTTP response message content includes the information as specified in R1AP [4], clause 7.3.4.2.

10.2.2.2.4 Expected result

The test is considered passed if Step 6 validation has passed.

10.2.2.3 Query data job (positive case)

10.2.2.3.1 Test description and applicability

The purpose of this test case is to test the query data job API as specified in R1AP [4], clause 7.3.4.5 The expected outcome is successful validation of the request from the DUT.

10.2.2.3.2 Test entrance criteria

1) The test simulator has functionality to initiate the query data subscription procedure as defined in R1GAP [1], clause 5.2.5.

2) A data job exists in the DUT and the dataJobId is known to the test simulator.

10.2.2.3.3 Test methodology

10.2.2.3.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP requests from the DUT.

10.2.2.3.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to query a data job.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.3.5.3.

b) The HTTP request is a GET operation.

c) The dataJobID in the URI match the data job being queried.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 7.3.4.5.1.

Step 4. The DUT constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 7.3.4.5.1.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.3.5.3.

b) The HTTP response message content includes the information as specified in R1AP [4], clause 7.3.4.5.1.

10.2.2.3.4 Expected result

The test is considered passed if Step 6 validation has passed.

10.2.2.4 Query data job status (positive case)

10.2.2.4.1 Test description and applicability

The purpose of this test case is to test the query data job status API as specified in R1AP [4], clause 7.3.4.6 The expected outcome is successful validation of the request from the DUT.

10.2.2.4.2 Test entrance criteria

1) The test simulator has functionality to initiate the query data subscription status procedure as defined in R1GAP [1], clause 5.2.5.

2) A data job exists in the DUT and the dataJobId is known to the test simulator.

10.2.2.4.3 Test methodology

10.2.2.4.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP requests from the DUT.

10.2.2.4.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to query a data job status.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.3.5.4.

b) The HTTP request is a GET operation.

c) The dataJobID in the URI match the data job status being queried.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 7.3.4.6.1.

Step 4. The DUT constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 7.3.4.6.1.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.3.5.4.

b) The HTTP response message content includes the information as specified in R1AP [4], clause 7.3.4.6.1.

10.2.2.4.4 Expected result

The test is considered passed if Step 6 validation has passed.

10.2.2.5 Query data job identifiers (positive case)

10.2.2.5.1 Test description and applicability

The purpose of this test case is to test the query data job identifiers API as specified in R1AP [4], clause 7.3.4.7 The expected outcome is successful validation of the request from the DUT.

10.2.2.5.2 Test entrance criteria

1) The test simulator has functionality to initiate the query data subscription status procedure as defined in R1GAP [1], clause 5.2.5.

2) A set of data jobs exists in the DUT.

10.2.2.5.3 Test methodology

10.2.2.5.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP requests from the DUT.

10.2.2.5.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to query a set of data job identifiers.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.3.5.2.

b) The HTTP request is a GET operation.

c) The HTTP request message content includes the information as specified in R1AP [4], clause 7.3.4.7.1.

Step 4. The DUT constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 7.3.4.7.1.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.3.5.2.

b) The HTTP response message content includes the information as specified in R1AP [4], clause 7.3.4.7.1.

10.2.2.5.4 Expected result

The test is considered passed if Step 6 validation has passed.

10.2.2.6 Update data job (positive case)

10.2.2.6.1 Test description and applicability

The purpose of this test case is to test the Update data job API as specified in R1AP [4], clause 7.3.4.4. The expected outcome is successful validation of the request from the DUT.

10.2.2.6.2 Test entrance criteria

1) The test simulator has functionality to initiate the Update data job procedure as defined in R1GAP [1], clause 5.2.5.
2) A data job exists in the DUT and the dataJobId and DataJobInfo are known to the test simulator.

10.2.2.6.3 Test methodology

10.2.2.6.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

10.2.2.6.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to update a data job identified by the dataJobId and DataJobInfo.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.3.5.3.
b) The HTTP request is a PUT operation.
c) The dataJobId and DataJobInfo in the URI match the data job being updated
d) The HTTP request message content includes the information as specified in R1AP [4], clause 7.3.4.4

Step 4. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 7.3.5.3.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

e) The URI conforms to the format specified in R1AP [4], clause 7.3.5.3.
f) The HTTP response message content includes the information as specified in R1AP [4], clause 7.3.4.4.

10.2.2.6.4 Expected result

The test is considered passed if Step 6 validation has passed.

10.2.2.7 Notify data availability (positive case)

10.2.2.7.1 Test description and applicability

The purpose of this test case is to test the Notify data availability API as specified in R1AP [4], clause 7.3.4.3. The expected outcome is successful validation of the request from the DUT.

10.2.2.7.2 Test entrance criteria

1) The test simulator has functionality to initiate the Notify data availability procedure as defined in R1GAP [1], clause 5.2.5.
2) A data subscription exists in the DUT.

10.2.2.7.3 Test methodology

10.2.2.7.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP requests from the DUT.

10.2.2.7.3.2 Procedure

Step 1. The DUT as an API producer initiates the sending of a HTTP POST request.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The HTTP request is a POST operation.

b) The callback URI matches the URI provided in the data subscription.

10.2.2.7.4 Expected result

The test is considered passed if Step 3 validation has passed.

11 A1 related services test cases

11.1 Conformance test cases for rApp

11.1.1 General

11.1.1.1 Device under test requirements

The rApp that acts as DUT in these test scenarios, the requirements on the DUT for these tests are that it can handle the A1 policy management service, and the purpose of the test scenarios is to validate that it conforms to the API Consumer functionality as specified in R1AP [4], clause 9.1.4.

11.1.1.2 Test simulator capabilities

The test simulator has the capabilities as required for a SMO/Non-RT RIC framework. In addition, it has the following capabilities:

- Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.
- Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT’s conformance to the API Consumer functionality as specified in R1AP [4], clause 9.1.4.

11.1.2 A1 policy management API as API Consumer test scenario

11.1.2.1 Query A1 policy type identifiers (positive case)

11.1.2.1.1 Test description and applicability

The purpose of this test case is to test the query A1 policy type identifiers API as specified in R1AP [4], clause 9.1.4.1.The expected outcome is successful validation of the request from the DUT.

11.1.2.1.2 Test entrance criteria

1) The DUT has functionality to initiate the Query A1 policy type identifiers procedure as defined in R1GAP [1], clause 5.3.2.
2) A set of A1 policy types exists in test simulator.

11.1.2.1.3 Test methodology

11.1.2.1.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

11.1.2.1.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to query A1 policy type identifiers.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.4.1.
b) The HTTP request is a GET operation.
c) The HTTP request message content includes the information as specified in R1AP [4], clause 9.1.4.1.

Step 4. The test simulator constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 9.1.5.2.

11.1.2.1.4 Expected result

The test is considered passed if Step 3 validation has passed.

11.1.2.2 Query A1 policy type (positive case)

11.1.2.2.1 Test description and applicability

The purpose of this test case is to test the query A1 policy type API as specified in R1AP [4], clause 9.1.4.2 The expected outcome is successful validation of the request from the DUT.

11.1.2.2.2 Test entrance criteria

1) The DUT has functionality to initiate the query A1 policy type procedure as defined in R1GAP [1], clause 5.3.2.
2) An A1 policy type exists in test simulator and the policyTypeId is known to the DUT.

11.1.2.2.3 Test methodology

11.1.2.2.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

11.1.2.2.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to query an A1 policy type.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.4.2.

b) The HTTP request is a GET operation.

c) The policyTypeID in the URI match the A1 policy type being queried. The HTTP request message content includes the information as specified in R1AP [4], clause 9.1.4.2.

Step 4. The test simulator constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 9.1.5.3.

11.1.2.2.4 Expected result

The test is considered passed if Step 3 validation has passed.

11.1.2.3 Query A1 policy identifiers (positive case)

11.1.2.3.1 Test description and applicability

The purpose of this test case is to test the query A1 policy identifiers API as specified in R1AP [4], clause 9.1.4.3 The expected outcome is successful validation of the request from the DUT.

11.1.2.3.2 Test entrance criteria

1) The DUT has functionality to initiate the query A1 policy identifiers procedure as defined in R1GAP [1], clause 5.3.2.

2) A set of A1 policies exists in test simulator.

11.1.2.3.3 Test methodology

11.1.2.3.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

11.1.2.3.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to query A1 policy identifiers.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.4.3.

b) The HTTP request is a GET operation.

c) The HTTP request message content includes the information as specified in R1AP [4], clause 9.1.4.3.

Step 4. The test simulator constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 9.1.5.4.

11.1.2.3.4 Expected result

The test is considered passed if Step 3 validation has passed.

11.1.2.4 Create A1 policy (positive case)

11.1.2.4.1 Test description and applicability

The purpose of this test case is to test the create A1 policy API as specified in R1AP [4], clause 9.1.4.4. The expected outcome is successful validation of the request from the DUT.

11.1.2.4.2 Test entrance criteria

1) The DUT has functionality to initiate the create A1 policy procedure as defined in R1GAP [1], clause 5.3.2.
2) The PolicyObjectInformation is known to the DUT.

11.1.2.4.3 Test methodology

11.1.2.4.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

11.1.2.4.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to create an A1 policy.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.4.4.
b) The HTTP request is a POST operation.
c) The HTTP request message content includes the information as specified in R1AP [4], clause 9.1.4.4.

Step 4. The test simulator constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 9.1.5.4.

11.1.2.4.4 Expected result

The test is considered passed if Step 3 validation has passed.

11.1.2.5 Query A1 policy (positive case)

11.1.2.5.1 Test description and applicability

The purpose of this test case is to test the query A1 policy API as specified in R1AP [4], clause 9.1.4.5 The expected outcome is successful validation of the request from the DUT.

11.1.2.5.2 Test entrance criteria

1) The DUT has functionality to initiate the query A1 policy procedure as defined in R1GAP [1], clause 5.3.2.
2) An A1 policy exists in test simulator and the policyId is known to the DUT.

11.1.2.5.3 Test methodology

11.1.2.5.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

11.1.2.5.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to query an A1 policy.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.4.5.

b) The HTTP request is a GET operation.

c) The policyId in the URI match the A1 policy being queried. The HTTP request message content includes the information as specified in R1AP [4], clause 9.1.4.5.

Step 4. The test simulator constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 9.1.5.5.

11.1.2.5.4 Expected result

The test is considered passed if Step 3 validation has passed.

11.1.2.6 Update A1 policy (positive case)

11.1.2.6.1 Test description and applicability

The purpose of this test case is to test the Update A1 policy API as specified in R1AP [4], clause 9.1.4.6. The expected outcome is successful validation of the request from the DUT.

11.1.2.6.2 Test entrance criteria

1) The DUT has functionality to initiate the Update A1 policy procedure as defined in R1GAP [1], clause 5.3.2.

2) An A1 policy exists in test simulator and the policyId and PolicyObject are known to the DUT.

11.1.2.6.3 Test methodology

11.1.2.6.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

11.1.2.6.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to update an A1 policy identified by the policyId and PolicyObject.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 9.1.4.6.

b) The HTTP request is a PUT operation.

c) The policyId and PolicyObject in the URI match the A1 policy being updated

d) The HTTP request message content includes the information as specified in R1AP [4], clause 9.1.4.6.

Step 4. The test simulator updates the resource, and a representation of the updated resource shall be returned in the response body the appropriate HTTP response as specified in R1AP [4], clause 9.1.5.5.

11.1.2.6.4 Expected result

The test is considered passed if Step 3 validation has passed

11.1.2.7 Delete A1 policy (positive case)

11.1.2.7.1 Test description and applicability

The purpose of this test case is to test the Delete A1 policy API as specified in R1AP [4], clause 9.1.4.6. The expected outcome is successful validation of the request from the DUT.

11.1.2.7.2 Test entrance criteria

1) The DUT has functionality to initiate the Delete A1 policy procedure as defined in R1GAP [1], clause 5.3.2.

2) An A1 policy exists in test simulator and the policyId is known to the DUT.

11.1.2.7.3 Test methodology

11.1.2.7.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

11.1.2.7.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to delete an A1 policy.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.4.7.

b) The HTTP request is a DELETE operation.

c) The policyId in the URI match the A1 policy being deleted. The HTTP request message content includes the information as specified in R1AP [4], clause 9.1.4.7.

Step 4. The test simulator constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4] , clause 9.1.5.5.

11.1.2.7.4 Expected result

The test is considered passed if Step 3 validation has passed.

11.1.2.8 Query A1 policy status (positive case)

11.1.2.8.1 Test description and applicability

The purpose of this test case is to test the query A1 policy status API as specified in R1AP [4], clause 9.1.4.8 The expected outcome is successful validation of the request from the DUT.

11.1.2.8.2 Test entrance criteria

1) The DUT has functionality to initiate the query A1 policy status procedure as defined in R1GAP [1], clause 5.3.2.
2) An A1 policy exists in test simulator and the policyId is known to the DUT.

11.1.2.8.3 Test methodology

11.1.2.8.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

11.1.2.8.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to query an A1 policy status.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.4.8.
b) The HTTP request is a GET operation.
c) The policyId in the URI match the A1 policy status being queried.
d) The HTTP request message content includes the information as specified in R1AP [4], clause 9.1.4.8 .

Step 4. The test simulator constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 9.1.5.6.

11.1.2.8.4 Expected result

The test is considered passed if Step 3 validation has passed.

11.1.2.9 Subscribe A1 policy status (positive case)

11.1.2.9.1 Test description and applicability

The purpose of this test case is to test the subscribe A1 policy status API as specified in R1AP [4], clause 9.1.4.9 The expected outcome is successful validation of the request from the DUT.

11.1.2.9.2 Test entrance criteria

1) The DUT has functionality to initiate the subscribe A1 policy status procedure as defined in R1GAP [1], clause 5.3.2.
2) The PolicyStatusSubscription is known to the DUT

11.1.2.9.3 Test methodology

11.1.2.9.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

11.1.2.9.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to create an A1 policy status subscription.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.4.9.

b) The HTTP request is a POST operation.

c) The HTTP request message content includes the information as specified in R1AP [4], clause 9.1.4.9.

Step 4. The test simulator constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 9.1.5.7.

11.1.2.9.4 Expected result

The test is considered passed if Step 3 validation has passed.

11.1.2.10 Update A1 policy status subscription (positive case)

11.1.2.10.1 Test description and applicability

The purpose of this test case is to test the Update A1 policy status subscription API as specified in R1AP [4], clause 9.1.4.10. The expected outcome is successful validation of the request from the DUT.

11.1.2.10.2 Test entrance criteria

1) The DUT has functionality to initiate the Update A1 policy status subscription procedure as defined in R1GAP [1], clause 5.3.2.

2) An A1 policy status subscription exists in test simulator and the subscriptionId and PolicyStatusSubscription are known to the DUT.

11.1.2.10.3 Test methodology

11.1.2.10.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

11.1.2.10.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to update an A1 policy status subscription identified by the subscriptionId and PolicyStatusSubscription.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 9.1.4.10.

b) The HTTP request is a PUT operation.

c) The subscriptionId and PolicyStatusSubscription in the URI match the status subscription being updated

d) The HTTP request message content includes the information as specified in R1AP [4], clause 9.1.4.10.

Step 4. The test simulator updates the resource, and a representation of the updated resource shall be returned in the response body the appropriate HTTP response as specified in R1AP [4], clause 9.1.5.8.

11.1.2.10.4 Expected result

The test is considered passed if Step 3 validation has passed

11.1.2.11 Query A1 policy status subscription (positive case)

11.1.2.11.1 Test description and applicability

The purpose of this test case is to test the query A1 policy status subscription API as specified in R1AP [4], clause 9.1.4.11 The expected outcome is successful validation of the request from the DUT.

11.1.2.11.2 Test entrance criteria

1) The DUT has functionality to initiate the query A1 policy status subscription procedure as defined in R1GAP [1], clause 5.3.2.

2) An A1 policy status subscription exists in test simulator and the subscriptionId is known to the DUT.

11.1.2.11.3 Test methodology

11.1.2.11.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

11.1.2.11.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to query an A1 policy status subscription.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.4.11.

b) The HTTP request is a GET operation.

c) The subscriptionId in the URI match the A1 policy status subscription being queried.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 9.1.4.11 .

Step 4. The test simulator constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 9.1.5.8.

11.1.2.11.4 Expected result

The test is considered passed if Step 3 validation has passed.

11.1.2.12 Unsubscribe A1 policy status (positive case)

11.1.2.12.1 Test description and applicability

The purpose of this test case is to test the Unsubscribe A1 policy status API as specified in R1AP [4], clause 9.1.4.12. The expected outcome is successful validation of the request from the DUT.

11.1.2.12.2 Test entrance criteria

1) The DUT has functionality to initiate the Unsubscribe A1 policy procedure as defined in R1GAP [1], clause 5.3.2.

2) An A1 policy status subscription exists in test simulator and the subscriptionId is known to the DUT.

11.1.2.12.3 Test methodology

11.1.2.12.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

11.1.2.12.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to delete an A1 policy status subscription.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.4.12.

b) The HTTP request is a DELETE operation.

c) The subscriptionId in the URI match the A1 policy status subscription being deleted. The HTTP request message content includes the information as specified in R1AP [4], clause 9.1.4.12.

Step 4. The test simulator constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 9.1.5.8.

11.1.2.12.4 Expected result

The test is considered passed if Step 3 validation has passed.

11.2 Conformance test cases for SMO/Non-RT RIC framework

11.2.1 General

11.2.1.1 Device under test requirements

The SMO/Non-RT RIC framework that acts as DUT in these test scenarios, the requirements on the DUT for these tests are that it can handle the DME Data subscription service, and the purpose of the test scenarios is to validate that it conforms to the API Producer functionality as specified in R1AP [4], clause 9.1.4.

11.2.1.2 Test simulator capabilities

The test simulator has the capabilities as required for a rApp. In addition, it has the following capabilities:

- Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.
- Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT’s conformance to the API Producer functionality as specified in R1AP [4], clause 9.1.4.

11.2.2 A1 policy management API as API Producer test scenario

11.2.2.1 Query A1 policy type identifiers (positive case)

11.2.2.1.1 Test description and applicability

The purpose of this test case is to test the query A1 policy type identifiers API as specified in R1AP [4], clause 9.1.4.1. The expected outcome is successful validation of the request from the DUT.

11.2.2.1.2 Test entrance criteria

1) The test simulator has functionality to initiate the Query A1 policy type identifiers procedure as defined in R1GAP [1], clause 5.3.2.

2) A set of A1 policy types exists in the DUT.

11.2.2.1.3 Test methodology

11.2.2.1.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP requests from the DUT.

11.2.2.1.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to query a set of A1 policy type identifiers.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.4.1.

b) The HTTP request is a GET operation.

c) The HTTP request message content includes the information as specified in R1AP [4], clause 9.1.4.1.

Step 4. The DUT constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 9.1.4.1.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.5.2.

b) The HTTP response message content includes the information as specified in R1AP [4], clause 9.1.4.1

11.2.2.1.4 Expected result

The test is considered passed if Step 6 validation has passed.

11.2.2.2 Query A1 policy type (positive case)

11.2.2.2.1 Test description and applicability

The purpose of this test case is to test the query A1 policy type API as specified in R1AP [4], clause 9.1.4.2 The expected outcome is successful validation of the request from the DUT.

11.2.2.2.2 Test entrance criteria

1) The DUT has functionality to initiate the query A1 policy type procedure as defined in R1GAP [1], clause 5.3.2.
2) An A1 policy type exists in DUT and the policyTypeId is known to the test simulator.

11.2.2.2.3 Test methodology

11.2.2.2.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP requests from the DUT.

11.2.2.2.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to query an A1 policy type.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.5.3.
b) The HTTP request is a GET operation.
c) The policyTypeId in the URI match the A1 policy type being queried. The HTTP request message content includes the information as specified in R1AP [4], clause 9.1.4.2.

Step 4. The DUT constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 9.1.4.2.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.5.3.
b) The HTTP response message content includes the information as specified in R1AP [4], clause 9.1.4.2.

11.2.2.2.4 Expected result

The test is considered passed if Step 6 validation has passed.

11.2.2.3 Query A1 policy identifiers (positive case)

11.2.2.3.1 Test description and applicability

The purpose of this test case is to test the query A1 policy identifiers API as specified in R1AP [4], clause 9.1.4.3. The expected outcome is successful validation of the request from the DUT.

11.2.2.3.2 Test entrance criteria

1) The test simulator has functionality to initiate the Query A1 policy identifiers procedure as defined in R1GAP [1], clause 5.3.2.
2) A set of A1 policies exists in the DUT.

11.2.2.3.3 Test methodology

11.2.2.3.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP requests from the DUT.

11.2.2.3.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to query a set of A1 policy identifiers.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.5.4.

b) The HTTP request is a GET operation.

c) The HTTP request message content includes the information as specified in R1AP [4], clause 9.1.4.3.

Step 4. The DUT constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 9.1.4.3.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.5.4.

b) The HTTP response message content includes the information as specified in R1AP [4], clause 9.1.4.3.

11.2.2.3.4 Expected result

The test is considered passed if Step 6 validation has passed.

11.2.2.4 Create A1 policy (positive case)

11.2.2.4.1 Test description and applicability

The purpose of this test case is to test the create A1 policy API as specified in R1AP [4], clause 9.1.4.4. The expected outcome is successful validation of the request from the DUT.

11.2.2.4.2 Test entrance criteria

1) The test simulator has functionality to initiate the create A1 policy procedure as defined in R1GAP [1], clause 5.3.2.

2) The PolicyObjectInformation is known to the test simulator

11.2.2.4.3 Test methodology

11.2.2.4.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

11.2.2.4.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to create an A1 policy.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The HTTP request is a POST request.

b) The URI conforms to the format specified in R1AP [4], clause 9.1.5.4

c) The HTTP request message content includes the information as specified in R1AP [4], clause 9.1.4.4.

Step 4. The DUT generates the A1 policy and constructs the URI for the created resource.

Step 5. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 9.1.4.4.

Step 6. At the test simulator the contents of the received HTTP response is recorded.

Step 7. The test simulator does the following validation:

a) The HTTP response message content includes the information as specified in R1AP [4], clause 9.1.4.4, and the Location header will be present.

b) The URI conforms to the format specified in R1AP [4], clause 9.1.5.4.

11.2.2.4.4 Expected result

The test is considered passed if Step 7 validation has passed.

11.2.2.5 Query A1 policy (positive case)

11.2.2.5.1 Test description and applicability

The purpose of this test case is to test the query A1 policy API as specified in R1AP [4], clause 9.1.4.5 The expected outcome is successful validation of the request from the DUT.

11.2.2.5.2 Test entrance criteria

1) The test simulator has functionality to initiate the query A1 policy procedure as defined in R1GAP [1], clause 5.3.2.

2) An A1 policy exists in DUT and the policyId is known to the test simulator.

11.2.2.5.3 Test methodology

11.2.2.5.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP requests from the DUT.

11.2.2.5.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to query an A1 policy.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.5.5.

b) The HTTP request is a GET operation.

c) The policyId in the URI match the A1 policy being queried.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 9.1.4.5.

Step 4. The DUT constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 9.1.4.5.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.5.5

b) The HTTP response message content includes the information as specified in R1AP [4], clause 9.1.4.5.

11.2.2.5.4 Expected result

The test is considered passed if Step 6 validation has passed.

11.2.2.6 Update A1 policy (positive case)

11.2.2.6.1 Test description and applicability

The purpose of this test case is to test the Update A1 policy API as specified in R1AP [4], clause 9.1.4.6. The expected outcome is successful validation of the request from the DUT.

11.2.2.6.2 Test entrance criteria

1) The test simulator has functionality to initiate the Update A1 policy procedure as defined in R1GAP [1], clause 5.3.2.

2) An A1 policy exists in the DUT and the policyId and PolicyObject are known to the test simulator.

11.2.2.6.3 Test methodology

11.2.2.6.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP requests from the DUT.

11.2.2.6.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to update an A1 policy identified by the policyId and PolicyObject.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.5.5.

b) The HTTP request is a PUT operation.

c) The policyId and PolicyObject in the URI match the A1 policy being updated

d) The HTTP request message content includes the information as specified in R1AP [4], clause 9.1.4.6

Step 4. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 9.1.4.6.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.5.5.

b) The HTTP response message content includes the information as specified in R1AP [4], clause 9.1.4.6.

11.2.2.6.4 Expected result

The test is considered passed if Step 6 validation has passed.

11.2.2.7 Delete A1 policy (positive case)

11.2.2.7.1 Test description and applicability

The purpose of this test case is to test the Delete A1 policy API as specified in R1AP [4], clause 9.1.4.6. The expected outcome is successful validation of the request from the DUT.

11.2.2.7.2 Test entrance criteria

1) The test simulator has functionality to initiate the Delete A1 policy procedure as defined in R1GAP [1], clause 5.3.2.
2) An A1 policy exists in DUT and the policyId is known to the test simulator.

11.2.2.7.3 Test methodology

11.2.2.7.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

11.2.2.7.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to Delete an A1 policy

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.5.5.
b) The HTTP request is a DELETE operation.
c) The policyID in the URI match the A1 policy being deleted.
d) The HTTP request message content includes the information as specified in R1AP [4], clause 9.1.4.7

Step 4. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 9.1.4.7.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 9.1.5.5.
b) The HTTP response message content includes the information as specified in R1AP [4], clause 9.1.4.7

11.2.2.7.4 Expected result

The test is considered passed if Step 6 validation has passed.

11.2.2.8 Query A1 policy status (positive case)

11.2.2.8.1 Test description and applicability

The purpose of this test case is to test the query A1 policy status API as specified in R1AP [4], clause 9.1.4.8 The expected outcome is successful validation of the request from the DUT.

11.2.2.8.2 Test entrance criteria

1) The DUT has functionality to initiate the query A1 policy procedure as defined in R1GAP [1], clause 5.3.2.

2) An A1 policy exists in DUT and the policyId is known to the test simulator.

11.2.2.8.3 Test methodology

11.2.2.8.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP requests from the DUT.

11.2.2.8.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to query an A1 policy status.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.5.36

b) The HTTP request is a GET operation.

c) The policyId in the URI match the A1 policy status being queried.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 9.1.4.8.

Step 4. The DUT constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 9.1.4.8.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.5.6.

b) The HTTP response message content includes the information as specified in R1AP [4], clause 9.1.4.8.

11.2.2.8.4 Expected result

The test is considered passed if Step 6 validation has passed.

11.2.2.9 Subscribe A1 policy status (positive case)

11.2.2.9.1 Test description and applicability

The purpose of this test case is to test the subscribe A1 policy status API as specified in R1AP [4], clause 9.1.4.9 The expected outcome is successful validation of the request from the DUT.

11.2.2.9.2 Test entrance criteria

1) The test simulator has functionality to initiate the subscribe A1 policy status procedure as defined in R1GAP [1], clause 5.3.2.

2) The PolicyStatusSubscription is known to the test simulator

11.2.2.9.3 Test methodology

11.2.2.9.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP requests from the DUT.

11.2.2.9.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to create an A1 policy status subscription.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The DUT does the following validation:

a) The HTTP request is a POST request.

b) The URI conforms to the format specified in R1AP [4], clause 9.1.5.7

c) The HTTP request message content includes the information as specified in R1AP [4], clause 9.1.4.9.

Step 4. The DUT generates the A1 policy status subscription and constructs the URI for the created resource.

Step 5. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 9.1.4.9.

Step 6. At the test simulator the contents of the received HTTP response is recorded.

Step 7. The test simulator does the following validation:

a) The HTTP response message content includes the information as specified in R1AP [4], clause 9.1.4.9, and the Location header will be present.

b) The URI conforms to the format specified in R1AP [4], clause 9.1.5.7.

11.2.2.9.4 Expected result

The test is considered passed if Step 7 validation has passed.

11.2.2.10 Update A1 policy status subscription (positive case)

11.2.2.10.1 Test description and applicability

The purpose of this test case is to test the Update A1 policy status subscription API as specified in R1AP [4], clause 9.1.4.10. The expected outcome is successful validation of the request from the DUT.

11.2.2.10.2 Test entrance criteria

1) The test simulator has functionality to initiate the Update A1 policy status subscription procedure as defined in R1GAP [1], clause 5.3.2.

2) An A1 policy status subscription exists in the DUT and the subscriptionId and PolicyStatusSubscription are known to the test simulator.

11.2.2.10.3 Test methodology

11.2.2.10.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP requests from the DUT.

11.2.2.10.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to update an A1 policy status subscription identified by the subscriptionId and PolicyStatusSubscription.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.5.8.

b) The HTTP request is a PUT operation.

c) The subscriptionId and PolicyStatusSubscription in the URI match the A1 policy status subscription being updated

d) The HTTP request message content includes the information as specified in R1AP [4], clause 9.1.4.10

Step 4. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 9.1.4.10.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.5.8.

b) The HTTP response message content includes the information as specified in R1AP [4], clause 9.1.4.10.

11.2.2.10.4 Expected result

The test is considered passed if Step 6 validation has passed

11.2.2.11 Query A1 policy status subscription (positive case)

11.2.2.11.1 Test description and applicability

The purpose of this test case is to test the query A1 policy status subscription API as specified in R1AP [4], clause 9.1.4.11 The expected outcome is successful validation of the request from the DUT.

11.2.2.11.2 Test entrance criteria

1) The DUT has functionality to initiate the query A1 policy status subscription procedure as defined in R1GAP [1], clause 5.3.2.

2) An A1 policy status subscription exists in test simulator and the subscriptionId is known to the DUT.

11.2.2.11.3 Test methodology

11.2.2.11.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP requests from the DUT.

11.2.2.11.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to query an A1 policy status subscription.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.5.8 .

b) The HTTP request is a GET operation.

c) The subscriptionId in the URI match the A1 policy status subscription being queried.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 9.1.4.11.

Step 4. The DUT constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 9.1.4.5.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.5.8

b) The HTTP response message content includes the information as specified in R1AP [4], clause 9.1.4.11.

11.2.2.11.4 Expected result

The test is considered passed if Step 6 validation has passed.

11.2.2.12 Unsubscribe A1 policy status (positive case)

11.2.2.12.1 Test description and applicability

The purpose of this test case is to test the Unsubscribe A1 policy status API as specified in R1AP [4], clause 9.1.4.12. The expected outcome is successful validation of the request from the DUT.

11.2.2.12.2 Test entrance criteria

1) The DUT has functionality to initiate the Unsubscribe A1 policy procedure as defined in R1GAP [1], clause 5.3.2.

2) An A1 policy subscription exists in DUT and the subscriptionId is known to the test simulator.

11.2.2.12.3 Test methodology

11.2.2.12.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

11.2.2.12.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to delete an A1 policy status subscription

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 9.1.5.8.

b) The HTTP request is a DELETE operation.

c) The subscriptionId in the URI match the A1 policy status subscription being deleted.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 9.1.4.12.

Step 4. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 9.1.4.12.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 9.1.5.8.

b) The HTTP response message content includes the information as specified in R1AP [4], clause 9.1.4.12.

11.2.2.12.4 Expected result

The test is considered passed if Step 6 validation has passed.

11.2.2.13 Notify A1 policy status changes (positive case)

11.2.2.13.1 Test description and applicability

The purpose of this test case is to test the Notify A1 policy status changes API as specified in R1AP [4], clause 9.1.4.13. The expected outcome is successful validation of the request from the DUT.

11.2.2.13.2 Test entrance criteria

1) The test simulator has functionality to initiate the Notify A1 policy status changes procedure as defined in R1GAP [1], clause 5.3.2.
2) An A1 policy status exists in the DUT.

11.2.2.13.3 Test methodology

11.2.2.13.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP requests from the DUT.

11.2.2.13.3.2 Procedure

Step 1. The DUT as an API producer initiates the sending of a HTTP POST request.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The HTTP request is a POST operation.
b) The callback URI matches the URI provided in the A1 policy status subscription.

11.2.2.13.4 Expected result

The test is considered passed if Step 3 validation has passed.

12 DME Data discovery services test cases

12.1 Conformance test cases for rApp

12.1.1 General

12.1.1.1 Device under test requirements

The rApp that acts as Device Under Test (DUT) in these test scenarios, the requirements on the DUT for these tests are that it can handle the data discovery service, and the purpose of the test scenarios is to validate that it conforms to the data discovery API specified in R1AP [4], clause 7.2.4.

12.1.1.2 Test simulator capabilities

The test simulator has the capabilities as required for a SMO/Non-RT RIC framework. In addition, it has the following capabilities:

- Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.

- Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT's conformance API Producer functionality as specified in R1AP [4], clause 7.2.4.

12.1.2 Data discover API as API Consumer test scenario

12.1.2.1 Discover DME types (positive case)

12.1.2.1.1 Test description and applicability

The purpose of this test case is to test the Discover DME types operation as specified in R1AP [4], clause 7.2.4.1. The expected outcome is successful validation of the request from the DUT.

12.1.2.1.2 Test entrance criteria

1) The DUT has functionality to initiate the Discover DME types procedure.
2) A set of DMETypeRelatedCapabilities exist in the test simulator.

12.1.2.1.3 Test methodology

12.1.2.1.3.1 Initial conditions

1) The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

12.1.2.1.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to Discover DME types.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 7.2.5.2.

b) The HTTP request is a GET operation.

c) The message body is empty.

Step 4. The test simulator generated the appropriate HTTP response as specified in R1AP [4], clause 7.2.5.2.3.1

NOTE: Presence or validation of optional filter parameters is not used to determine validation on this test.

12.1.2.1.4 Expected result

The test is considered passed if Step 3 validation has passed.

12.1.3.1 Query capabilities related to DME type (positive case)

12.1.3.1.1 Test description and applicability

The purpose of this test case is to test the Query capabilities related to DME type operation as specified in R1AP [4], clause 7.2.4.2. The expected outcome is successful validation of the request from the DUT.

12.1.3.1.2 Test entrance criteria

1) The DUT has functionality to initiate the Query DME type information procedure.

2) A set of service DMETypeRelatedCapabilities exist in the test simulator.

12.1.3.1.3 Test methodology

12.1.3.1.3.1 Initial conditions

1) The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

12.1.3.1.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to query for information about a specific DME type with dmeTypeId.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 7.2.5.3.

b) The HTTP request is a GET operation.

c) The message body is empty.

Step 4. The test simulator generated the appropriate HTTP response as specified in R1AP [4], clause 7.2.5.3.3.1

12.1.3.1.4 Expected result

The test is considered passed if Step 3 validation has passed.

12.2 Conformance test cases for SMO/Non-RT RIC framework

12.2.1 General

12.2.1.1 Device under test requirements

The SMO/Non-RT RIC framework that acts as Device Under Test (DUT) in these test scenarios, the requirements on the DUT for these tests are that it can handle the SME service discovery service, and the purpose of the test scenarios is to validate that it conforms to the Data discovery API specified in R1AP [4], clause 7.2.4.

12.2.1.2 Test simulator capabilities

The test simulator has the capabilities as specified in section 4.3.2. In addition, it has the following capabilities:

1) Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.

2) Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT's conformance to API Producer functionality as specified in R1AP [4], clause 7.2.4.

12.2.2 data discovery API as API Producer test scenario

12.2.2.1 Discover DME types (positive case)

12.2.2.1.1 Test description and applicability

The purpose of this test case is to test the Data discovery APIs as specified in R1AP [4], clause 7.2. The expected outcome is successful validation of the response from the DUT.

12.2.2.1.2 Test entrance criteria

1) The DUT supports the functionality to Discover DME types procedure.
2) A set of DMETypeRelatedCapabilities are supported in the test simulator.

12.2.2.1.3 Test methodology

12.2.2.1.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

12.2.2.1.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to Data discovery APIs for all dme types .

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The HTTP request is a GET request.
b) The URI conforms to the format as specified in R1AP [4], clause 7.2.5.2
c) The HTTP request message content includes the information as specified in R1AP[4], clause 7.2.5.2.3.1.

Step 4. The DUT initiates a HTTP Response.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.2.5.2.
b) The HTTP response message content includes the information as specified in R1AP [4], clause 7.2.5.2.3.1.

12.2.2.1.4 Expected result

The test is considered passed if Step 6 validation has passed.

12.2.3.1 Query capabilities related to DME type (positive case)

12.2.3.1.1 Test description and applicability

The purpose of this test case is to test the Data discovery APIs as specified in R1AP [4], clause 7.2. The expected outcome is successful validation of the response from the DUT.

12.2.3.1.2 Test entrance criteria

1) The DUT supports the functionality to Discover DME types procedure.

2) A set of DMETypeRelatedCapabilities are supported in the test simulator.

12.2.3.1.3 Test methodology

12.2.3.1.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

12.2.3.1.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to Data discovery APIs for all dme type with dmeTypeId .

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The HTTP request is a GET request.

b) The URI conforms to the format as specified in R1AP [4], clause 7.2.5.3

c) The HTTP request message content includes the information as specified in R1AP [4], clause 7.2.5.3.3.1.

Step 4. The DUT initiates a HTTP Response.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.2.5.3.

b) The HTTP response message content includes the information as specified in R1AP [4], clause 7.2.5.3.3.1.

12.2.3.1.4 Expected result

The test is considered passed if Step 6 validation has passed.

13 DME Data offer service test cases

13.1 Conformance test cases for rApp

13.1.1 General

13.1.1.1 Device under test requirements

The rApp that acts as DUT in these test scenarios, the requirements on the DUT for these tests are that it can handle the DME Data offer service, and the purpose of the test scenarios is to validate that it conforms to the API Consumer functionality as specified in R1AP [4], clause 7.6.4.

13.1.1.2 Test simulator capabilities

The test simulator has the capabilities as required for a SMO/Non-RT RIC framework. In addition, it has the following capabilities:

- Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.

- Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT's conformance to the API Consumer functionality as specified in R1AP [4], clause 7.6.4.

13.1.2 Data offer API as API Consumer test scenario

13.1.2.1 Create data offer (positive case)

13.1.2.1.1 Test description and applicability

The purpose of this test case is to test the create data offer API as specified in R1AP [4], clause 7.6.4.1. The expected outcome is successful validation of the request from the DUT.

13.1.2.1.2 Test entrance criteria

1) The DUT has functionality to initiate the create data offer procedure as defined in R1GAP [1], clause 5.2.7.
2) The DataOfferInfo is known to the DUT

13.1.2.1.3 Test methodology

13.1.2.1.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

13.1.2.1.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to create a data offer.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.6.4.1.

b) The HTTP request is a POST operation.

c) The HTTP request message content includes the information as specified in R1AP [4], clause 7.6.4.1.

Step 4. The test simulator constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 7.6.4.1.1.

13.1.2.1.4 Expected result

The test is considered passed if Step 3 validation has passed.

13.1.2.2 Cancel data offer (positive case)

13.1.2.2.1 Test description and applicability

The purpose of this test case is to test the Cancel data offer API as specified in R1AP [4], clause 7.6.4.2. The expected outcome is successful validation of the request from the DUT.

13.1.2.2.2 Test entrance criteria

1) The DUT has functionality to initiate the terminate data offer procedure as defined in R1GAP [1], clause 5.2.7.

2) A data offer exists in test simulator and the dataOfferId is known to the DUT.

13.1.2.2.3 Test methodology

13.1.2.2.3.1 Initial conditions

1. The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.
2. The dataOfferId exists in the test simulator.

13.1.2.2.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to cancel a data offer.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.6.5.3.

b) The HTTP request is a DELETE operation.

c) The dataOfferId in the URI matches the data offer being deleted.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 7.6.4.2.

Step 4. The test simulator generates and sends the appropriate HTTP response as specified in R1AP[4], clause 7.6.4.2.1.

13.1.2.2.4 Expected result

The test is considered passed if Step 3 validation has passed.

13.2 Conformance test cases for SMO/Non-RT RIC framework

13.2.1 General

13.2.1.1 Device under test requirements

The SMO/Non-RT RIC framework that acts as DUT in these test scenarios, the requirements on the DUT for these tests are that it can handle the DME Data offer service, and the purpose of the test scenarios is to validate that it conforms to the API Producer functionality as specified in R1AP [4], clause 7.6.4.

13.2.1.2 Test simulator capabilities

The test simulator has the capabilities as required for a rApp. In addition, it has the following capabilities:

- Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.
- Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT’s conformance to the API Producer functionality as specified in R1AP [4], clause 7.6.4.

13.2.2 Data offer API as API Producer test scenario

13.2.2.1 Create data offer (positive case)

13.2.2.1.1 Test description and applicability

The purpose of this test case is to test the create data offer API as specified in R1AP [4], clause 7.6.4.1. The expected outcome is successful validation of the request from the DUT.

13.2.2.1.2 Test entrance criteria

1) The test simulator has functionality to initiate the create data offer procedure as defined in R1GAP [1], clause 5.2.7.

13.2.2.1.3 Test methodology

13.2.2.1.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

13.2.2.1.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to create a data offer.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The HTTP request is a POST request.

b) The URI conforms to the format specified in R1AP [4], clause 7.6.5.2.3.1.

c) The HTTP request message content includes the information as specified in R1AP [4], clause 7.6.4.1.

Step 4. The DUT generates the data offer and constructs the URI for the created resource.

Step 5. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 7.6.4.1.

Step 6. At the test simulator the contents of the received HTTP response is recorded.

Step 7. The test simulator does the following validation:

a) The HTTP response message content includes the information as specified in R1AP [4], clause 7.6.4.1, and the Location header will be present.

b) The URI conforms to the format specified in R1AP [4], clause 7.6.5.3.

13.2.2.1.4 Expected result

The test is considered passed if Step 7 validation has passed.

13.2.2.2 Cancel data offer (positive case)

13.2.2.2.1 Test description and applicability

The purpose of this test case is to test the Cancel data offer API as specified in R1AP [4], clause 7.6.4.2. The expected outcome is successful validation of the request from the DUT.

13.2.2.2.2 Test entrance criteria

1) The test simulator has functionality to initiate the cancel data request procedure as defined in R1GAP [1], clause 5.2.7.
2) A data offer exists in DUT and the dataOfferId is known to the test simulator.

13.2.2.2.3 Test methodology

13.2.2.2.3.1 Initial conditions

1) The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.
2) The dataOfferId is known to the DUT

13.2.2.2.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to cancel a data offer.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.6.5.3.3.1.
b) The HTTP request is a DELETE operation.
c) The dataOfferId in the URI match the data offer being deleted.
d) The HTTP request message content includes the information as specified in R1AP [4], clause 7.6.4.2.

Step 4. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 7.6.4.2.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 7.6.5.3.
b) The HTTP response message content includes the information as specified in R1AP [4], clause 7.6.4.2.

13.2.2.2.4 Expected result

The test is considered passed if Step 6 validation has passed.

13.2.2.3 Notify data offer termination (positive case)

13.2.2.3.1 Test description and applicability

The purpose of this test case is to test the Notify data offer termination API as specified in R1AP [4], clause 7.6.4.3. The expected outcome is successful validation of the request from the DUT.

13.2.2.3.2 Test entrance criteria

1) The test simulator has functionality to initiate the Notify data offer termination procedure as defined in R1GAP [1], clause 5.2.7.
2) A data offer exists in the DUT.

13.2.2.3.3 Test methodology

13.2.2.3.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP requests from the DUT.

13.2.2.3.3.2 Procedure

Step 1. The DUT as an API producer initiates the sending of a HTTP POST request.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The HTTP request is a POST operation.

b) The callback URI matches the URI provided in the data offer.

Step 4. The test simulator sends the HTTP response as specified in R1AP [4], clause 7.6.4.3.

13.2.2.3.4 Expected result

The test is considered passed if Step 3 validation has passed.

14 SME Boot strap service test cases

14.1 Conformance test cases for rApp

14.1.1 General

14.1.1.1 Device under test requirements

The rApp that acts as Device Under Test (DUT) in these test scenarios, the requirements on the DUT for these tests are that it can handle the Bootstrap service, and the purpose of the test scenarios is to validate that it confirms API Consumer functionality as specified in R1AP [4], clause 6.4.4.

14.1.1.2 Test simulator capabilities

The test simulator has the capabilities as required for a SMO/Non-RT RIC framework. In addition, it has the following capabilities:

- Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.
- Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT's conformance API Producer functionality as specified in R1AP [4], clause 6.4.4.

14.1.2 Bootstrap API as API Consumer test scenario

14.1.2.1 Query bootstrap information (positive case)

14.1.2.1.1 Test description and applicability

The purpose of this test case is to test the Bootstrap APIs operation as specified in R1AP [4], clause 6.4.4.1. The expected outcome is successful validation of the request from the DUT.

14.1.2.1.2 Test entrance criteria

1) The DUT has functionality to initiate the Discover bootstrap procedure.
2) A BootstrapInformation exist in the test simulator.

14.1.2.1.3 Test methodology

14.1.2.1.3.1 Initial conditions

1) The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

14.1.2.1.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to BootstrapAPIs.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 6.4.5.2.
b) The HTTP request is a GET operation.
c) The message body is empty.

Step 4. The test simulator generated the appropriate HTTP response as specified in R1AP [4], clause 6.4.5.2.3.1.

NOTE: Presence or validation of optional filter parameters is not used to determine validation on this test.

14.1.2.1.4 Expected result

The test is considered passed if Step 3 validation has passed.

14.2 Conformance test cases for SMO/Non-RT RIC framework

14.2.1 General

14.2.1.1 Device under test requirements

The SMO/Non-RT RIC framework that acts as Device Under Test (DUT) in these test scenarios, the requirements on the DUT for these tests are that it can handle the Bootstrap service, and the purpose of the test scenarios is to validate that it conforms to the Bootstrap API specified in R1AP [4], clause 6.4.4.

14.2.1.2 Test simulator capabilities

The test simulator has the capabilities as specified in section 4.3.2. In addition, it has the following capabilities:

1) Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.

2) Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT's conformance to API Producer functionality as specified in R1AP [4], clause 6.4.4.

14.2.2 Bootstrap API as API Producer test scenario

14.2.2.1 Query bootstrap information (positive case)

14.2.2.1.1 Test description and applicability

The purpose of this test case is to test the Bootstrap APIs as specified in R1AP [4], clause 6.4.4.1. The expected outcome is successful validation of the response from the DUT.

14.2.2.1.2 Test entrance criteria

1) The DUT has functionality to initiate the Discover bootstrap procedure.

2) A BootstrapInformation exist in the test simulator.

14.2.2.1.3 Test methodology

14.2.2.1.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

14.2.2.1.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to Bootstrap API for bootstrap information.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The HTTP request is a GET request.

b) The URI conforms to the format as specified in R1AP [4], clause 6.2.5.2.

c) The HTTP request message content includes the information as specified in R1AP [4], clause 6.2.5.2.

Step 4. The DUT initiates a HTTP Response.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 6.2.5.2.

b) The HTTP response message content includes the information as specified in R1AP [4], clause. 6.2.5.2.

14.2.2.1.4 Expected result

The test is considered passed if Step 6 validation has passed.

15 AI/ML workflow AI/ML model registration API test cases

15.1 Conformance test cases for rApp

15.1.1 General

15.1.1.1 Device under test requirements

The rApp that acts as DUT in these test scenarios, the requirements on the DUT for these tests are that it can handle the AI/ML model registration service, and the purpose of the test scenarios is to validate that it conforms to the API Consumer functionality as specified in R1AP [4], clause 10.1.4.

15.1.1.2 Test simulator capabilities

The test simulator has the capabilities as required for a SMO/Non-RT RIC framework. In addition, it has the following capabilities:

- Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.
- Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT's conformance to the API Consumer functionality as specified in R1AP [4], clause 10.1.4.

15.1.2 AI/ML model registration API as API consumer test scenario

15.1.2.1 Register model information (positive case)

15.1.2.1.1 Test description and applicability

The purpose of this test case is to test the Register AI/ML model information in AI/ML model registration API as specified in R1AP [4], clause10.1. The expected outcome is successful validation of the request from the DUT.

15.1.2.1.2 Test entrance criteria

The DUT has functionality to initiate the Register AI/ML model procedure as defined in R1GAP [1], clause 5.6.2.

NOTE: The DUT provides the ModelRelatedInformation as defined in R1AP [4], clause 10.1.4.

15.1.2.1.3 Test methodology

15.1.2.1.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

15.1.2.1.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to register an AI/ML model by providing ModelRelatedInformation.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 10.1.5.2.2.

b) The HTTP request is a POST operation.

c) The HTTP request message content includes ModelRelatedInformation as specified in R1AP [4], clause 10.1.4.

Step 4. The test simulator generates the registrationId and constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4], clause 10.1.5.2.3.1.

15.1.2.1.4 Expected result

The test is considered passed if Step 3 validation has passed.

15.1.2.2 Deregister model information (positive case)

15.1.2.2.1 Test description and applicability

The purpose of this test case is to test the Deregister the registered model information in AI/ML model registration API as specified in R1AP [4], clause 10.1.4. The expected outcome is successful validation of the request from the DUT.

15.1.2.2.2 Test entrance criteria

1) The DUT has functionality to initiate the Deregister AI/ML model procedure as defined in R1GAP [1], clause 5.6.2.2.

2) A ModelRelatedInformation exists in test simulator and the registrationId is known to the DUT.

15.1.2.2.3 Test methodology

15.1.2.2.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

15.1.2.2.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to Deregister an AI/ML model information.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 10.1.5.3.2.

b) The HTTP request is a DELETE operation.

c) The registrationId in the URI match the ModelRelatedInformation being deregistered.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 10.1.5.3.3.

Step 4. The test simulator generated the appropriate HTTP response as specified in R1AP [4], clause 10.1.5.3.3

15.1.2.2.4 Expected result

The test is considered passed if Step 3 validation has passed.

15.1.2.3 Query model information (positive case)

15.1.2.3.1 Test description and applicability

The purpose of this test case is to test the Query model registration information in AI/ML model registration API as specified in R1AP [4], clause 10.1.4.4. The expected outcome is successful validation of the request from the DUT.

15.1.2.3.2 Test entrance criteria

1) The DUT has functionality to initiate the query AI/ML model registration procedure as defined in R1GAP [1], clause 5.6.2.2.
2) A ModelRelatedInformation exists in test simulator and the registrationId is known to the DUT.

15.1.2.3.3 Test methodology

15.1.2.3.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

15.1.2.3.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to query a ModelRelatedInformation.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 10.1.5.3.2.

b) The HTTP request is a GET operation.

c) The registrationId in the URI match the AI/ML model being queried.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 10.1.5.3.3.

Step 4. The test simulator generated the appropriate HTTP response as specified in R1AP [4], clause 10.1.5.3.3.

15.1.2.3.4 Expected result

The test is considered passed if Step 3 validation has passed.

15.1.2.4 Update model information (positive case)

15.1.2.4.1 Test description and applicability

The purpose of this test case is to test the Update registered model information in AI/ML model API as specified in R1AP [4], clause 10.1.4.3. The expected outcome is successful validation of the request from the DUT.

15.1.2.4.2 Test entrance criteria

1) The DUT has functionality to initiate the Update AI/ML model registration procedure as defined in R1GAP [1], clause 5.6.2.2.
2) An AI/Ml model exists in test simulator and the registrationId and ModelRelatedInformation are known to the DUT.

15.1.2.4.3 Test methodology

15.1.2.4.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

15.1.2.4.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to update a registered AI/ML model identified by the registrationId and ModelRelatedInformation.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 10.1.5.3.2.

b) The HTTP request is a PUT operation.

c) The registrationId and ModelRelatedInformation in the URI match the AI/Ml model being updated.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 10.1.5.3.3.

Step 4. The test simulator updates the resource, and a representation of the updated resource shall be returned in the response body the appropriate HTTP response as specified in R1AP [4], clause 10.1.5.3.3.

15.1.2.4.4 Expected result

The test is considered passed if Step 3 validation has passed.

15.2 Conformance test cases for SMO/Non-RT RIC framework

15.2.1 General

15.2.1.1 Device under test requirements

The SMO/Non-RT RIC framework that acts as DUT in these test scenarios, the requirements on the DUT for these tests are that it can handle the AI/ML model registration service, and the purpose of the test scenarios is to validate that it conforms to the API Producer functionality as specified in R1AP [4], clause 10.1.4.

15.2.1.2 Test simulator capabilities

The test simulator has the capabilities as required for a rApp. In addition, it has the following capabilities:

- Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.
- Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT’s conformance to the API Producer functionality as specified in R1AP [4], clause 10.1.4.

15.2.2 AI/ML model registration API as API Producer test scenario

15.2.2.1 Register model information (positive case)

15.2.2.1.1 Test description and applicability

The purpose of this test case is to test the Register AI/ML model information in AI/ML model registration API as specified in R1AP [4], clause 10.1.4. The expected outcome is successful validation of the request from the DUT.

15.2.2.1.2 Test entrance criteria

The test simulator has functionality to initiate the Register AI/ML model procedure as defined in R1GAP [1], clause 5.6.2.

15.2.2.1.3 Test methodology

15.2.2.1.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

15.2.2.1.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to register a ModelRelatedInformation.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The HTTP request is a POST request.

b) The URI conforms to the format specified in R1AP [4], clause 10.1.5.2.2.

c) The HTTP request message content includes the information as specified in R1AP [4], clause 10.1.5.2.3.1.

Step 4. The DUT generates the registrationId and constructs the URI for the created resource.

Step 5. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 10.1.5.2.3.1.

Step 6. At the test simulator the contents of the received HTTP response is recorded.

Step 7. The test simulator does the following validation:

a) The HTTP response message content includes the information as specified in R1AP [4], clause 10.1.5.2.3.1, and the Location header will be present.

b) The URI conforms to the format specified in R1AP [4], clause 10.1.5.2.2.

15.2.2.1.4 Expected result

The test is considered passed if Step 7 validation has passed.

15.2.2.2 Deregister model information (positive case)

15.2.2.2.1 Test description and applicability

The purpose of this test case is to test the Deregister registered AI/ML model information in AI/ML model registration API as specified in R1AP [4], clause 10.1.4. The expected outcome is successful validation of the request from the DUT.

15.2.2.2.2 Test entrance criteria

1) The test simulator has functionality to initiate the AI/ML model procedure as defined in R1GAP [1], clause 5.6.2.2.

2) A ModelRelatedInformation exists in test simulator and the registrationId is known to the DUT.

15.2.2.2.3 Test methodology

15.2.2.2.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

15.2.2.2.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to Deregister an AI/ML model information.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 10.1.5.3.2.

b) The HTTP request is a DELETE operation.

c) The registrationId in the URI match the ModelRelatedInformation being deregistered.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 10.1.5.3.3.

Step 4. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 10.1.5.3.3.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 10.1.5.3.2.

b) The HTTP response message content includes the information as specified in R1AP [4], clause 10.1.5.3.3.

15.2.2.2.4 Expected result

The test is considered passed if Step 6 validation has passed.

15.2.2.3 Query model information (positive case)

15.2.2.3.1 Test description and applicability

The purpose of this test case is to test the Query model registration information in AI/ML model registration API as specified in R1AP [4], clause 10.1.4.4. The expected outcome is successful validation of the request from the DUT.

15.2.2.3.2 Test entrance criteria

1) The test simulator has functionality to initiate the query AI/ML model registration procedure as defined in R1GAP [1], clause 5.6.2.2.

2) A ModelRelatedInformation exists in test simulator and the registrationId is known to the DUT.

15.2.2.3.3 Test methodology

15.2.2.3.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

15.2.2.3.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to query a ModelRelatedInformation.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 10.1.5.3.2.

b) The HTTP request is a GET operation.

c) The registrationId in the URI match the AI/ML model being queried.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 10.1.5.3.3.

Step 4. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 10.1.5.3.3.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 10.1.5.3.2.

b) The HTTP response message content includes the information as specified in R1AP [4], clause 10.1.5.3.3.

15.2.2.3.4 Expected result

The test is considered passed if Step 6 validation has passed.

15.2.2.4 Update model information (positive case)

15.2.2.4.1 Test description and applicability

The purpose of this test case is to test the Update registered model information in AI/ML model API as specified in R1AP [4], clause 10.1.4.3. The expected outcome is successful validation of the request from the DUT.

15.2.2.4.2 Test entrance criteria

1) The test simulator has functionality to initiate the Update AI/ML model registration procedure as defined in R1GAP [1], clause 5.6.2.2

2) An AI/ML model exists in test simulator and the registrationId and ModelRelatedInformation are known to the DUT.

15.2.2.4.3 Test methodology

15.2.2.4.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

15.2.2.4.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to update an AI/ML model information identified by the registrationId and ModelRelatedInformation.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The HTTP request is a PUT request.

b) The URI conforms to the format specified in R1AP [4], clause 10.1.5.3.2.

c) The HTTP request message content includes the information as specified in R1AP [4], clause 10.1.5.3.3.

Step 4. The DUT generates the registrationId and constructs the URI for the created resource.

Step 5. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 10.1.5.3.3.

Step 6. At the test simulator the contents of the received HTTP response is recorded.

Step 7. The test simulator does the following validation:

a) The HTTP response message content includes the information as specified in R1AP [4], clause 10.1.5.3.3.

b) The URI conforms to the format specified in R1AP [4], clause 10.1.5.3.2.

15.2.2.4.4 Expected result

The test is considered passed if Step 7 validation has passed.

16 AI/ML workflow AI/ML model discovery API test cases

16.1 Conformance test cases for rApp

16.1.1 General

16.1.1.1 Device under test requirements

The rApp that acts as Device Under Test (DUT) in these test scenarios, the requirements on the DUT for these tests are that it can handle the AI/ML model discovery service, and the purpose of the test scenarios is to validate that it conforms to the AI/ML model discovery API specified in R1AP [4], clause 10.2.4.

16.1.1.2 Test simulator capabilities

The test simulator has the capabilities as required for a SMO/Non-RT RIC framework. In addition, it has the following capabilities:

- Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.
- Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT's conformance API Producer functionality as specified in R1AP [4], clause 10.2.4.

16.1.2 AI/ML model discovery API as API Consumer test scenario

16.1.2.1 Discover AI/ML models (positive case)

16.1.2.1.1 Test description and applicability

The purpose of this test case is to test the Discover registered AI/ML models as specified in R1AP [4], clause 10.2.4.1. The expected outcome is successful validation of the request from the DUT.

16.1.2.1.2 Test entrance criteria

1) The DUT has functionality to initiate the Discover AI/ML model procedure.
2) A set of ModelRelatedInformation exist in the test simulator.

16.1.2.1.3 Test methodology

16.1.2.1.3.1 Initial conditions

1) The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

16.1.2.1.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to Discover registered AI/ML models.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 10.2.5.2.2.

b) The HTTP request is a GET operation.

c) The message body is empty.

Step 4. The test simulator generated the appropriate HTTP response as specified in R1AP [4], clause 10.2.5.2.3.1

NOTE: Presence or validation of optional filter parameters is not used to determine validation on this test.

16.1.2.1.4 Expected result

The test is considered passed if Step 3 validation has passed.

16.1.2.2 Query AI/ML models (positive case)

16.1.2.2.1 Test description and applicability

The purpose of this test case is to test the Query AI/ML models API as specified in R1AP [4], clause 10.2.4.2. The expected outcome is successful validation of the request from the DUT.

16.1.2.2.2 Test entrance criteria

1) The DUT has functionality to initiate the Discover AI/ML model procedure as defined in R1GAP [1], clause 5.6.4.

2) A model exists in the test simulator and the modelId is known to the DUT

16.1.2.2.3 Test methodology

16.1.2.2.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

16.1.2.2.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to Query registered AI/ML models.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 10.2.5.3.2.

b) The HTTP request is a GET operation.

c) The modelId in the URI match the model being queried.

Step 4. The test simulator generated the appropriate HTTP response as specified in R1AP [4], clause 10.2.4.2

16.1.2.2.4 Expected result

The test is considered passed if Step 3 validation has passed.

16.2 Conformance test cases for SMO/Non-RT RIC framework

16.2.1 General

16.2.1.1 Device under test requirements

The SMO/Non-RT RIC framework that acts as Device Under Test (DUT) in these test scenarios, the requirements on the DUT for these tests are that it can handle the AI/ML model discovery service, and the purpose of the test scenarios is to validate that it conforms to the AI/ML model discovery API specified in R1AP [4], clause 10.2.4.

16.2.1.2 Test simulator capabilities

The test simulator has the capabilities as specified in section 4.3.2. In addition, it has the following capabilities:

1) Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.

2) Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT's conformance to API Producer functionality as specified in R1AP [4], clause 10.2.4.

16.2.2 AI/ML model discovery API as API Producer test scenario

16.2.2.1 Discover AI/ML models (positive case)

16.2.2.1.1 Test description and applicability

The purpose of this test case is to test the AI/ML model discovery APIs as specified in R1AP [4], clause 10.2.4. The expected outcome is successful validation of the response from the DUT.

16.2.2.1.2 Test entrance criteria

1) The DUT supports the functionality to Discover AI/ML model procedure.

2) A set of ModelRelatedInformation are supported in the test simulator.

16.2.2.1.3 Test methodology

16.2.2.1.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

16.2.2.1.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to AI/ML model discovery APIs for registered AI/ML models.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The HTTP request is a GET request.

b) The URI conforms to the format as specified in R1AP [4], clause 10.2.5.2.2

c) The HTTP request message content includes the information as specified in R1AP [4], clause 10.2.5.2.3.1.

Step 4. The DUT initiates a HTTP Response.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 10.2.5.2.2.

b) The HTTP response message content includes the information as specified in R1AP [4], clause 10.2.5.2.3.1.

16.2.2.1.4 Expected result

The test is considered passed if Step 6 validation has passed.

16.2.2.2 Query AI/ML models (positive case)

16.2.2.2.1 Test description and applicability

The purpose of this test case is to test the Query AI/ML model API as specified in R1AP [4], clause 10.2.4.2. The expected outcome is successful validation of the response from the DUT.

16.2.2.2.2 Test entrance criteria

1) The DUT supports the functionality to Discover AI/ML model procedure.

2) A model exists in the DUT and the modelId is known to the test simulator.

16.2.2.2.3 Test methodology

16.2.2.2.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

16.2.2.2.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to Query a registered AI/ML models.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The HTTP request is a GET request.

b) The URI conforms to the format as specified in R1AP [4], clause 10.2.5.3.2

c) The HTTP request message content includes the information as specified in R1AP [4], clause 10.2.4.2.

Step 4. The DUT initiates a HTTP Response.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 10.2.5.3.2.

b) The HTTP response message content includes the information as specified in R1AP [4], clause 10.2.4.2.

16.2.2.2.4 Expected result

The test is considered passed if Step 6 validation has passed.

17 RAN OAM Configuration management service test cases

17.1 Conformance test cases for rApp

17.1.1 General

17.1.1.1 Device under test requirements

The rApp that acts as DUT in these test scenarios, the requirements on the DUT for these tests are that it can handle the RAN OAM Configuration management service, and the purpose of the test scenarios is to validate that it conforms to the API Consumer functionality as specified in R1AP [4], clause 8.1.4.

17.1.1.2 Test simulator capabilities

The test simulator has the capabilities as required for a SMO/Non-RT RIC framework. In addition, it has the following capabilities:

- Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.
- Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT's conformance to the API Consumer functionality as specified in R1AP [4], clause 8.1.4.

17.1.2 Configuration management API as API Consumer test scenario

17.1.2.1 Read configuration data (positive case)

17.1.2.1.1 Test description and applicability

The purpose of this test case is to test the read configuration data API as specified in R1AP [4], clause 8.1.4.1. The expected outcome is successful validation of the request from the DUT.

17.1.2.1.2 Test entrance criteria

1) The DUT has functionality to initiate the read configuration data procedure as defined in R1GAP [1], clause 5.4.5.
2) A managed entity with configuration data exists in the test simulator and the URI parameter and className are known to the DUT

17.1.2.1.3 Test methodology

17.1.2.1.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

17.1.2.1.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to read configuration data.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 8.1.4.1.

b) The HTTP request is a GET operation.

c) The HTTP request message content includes the information as specified in R1AP [4], clause 8.1.4.1.

Step 4. The test simulator sends the appropriate HTTP response as specified in R1AP [4], clause 8.1.4.1.1.

17.1.2.1.4 Expected result

The test is considered passed if Step 3 validation has passed.

17.1.2.2 Write configuration changes (positive case)

17.1.2.2.1 Test description and applicability

The purpose of this test case is to test the Write configuration changes API as specified in R1AP [4], clause 8.1.4.2. The expected outcome is successful validation of the request from the DUT.

17.1.2.2.2 Test entrance criteria

1) The DUT has functionality to initiate the write configuration procedure as defined in R1GAP [1], clause 5.4.5.

2) A managed entity with configuration data exists in the test simulator and the URI parameter and className are known to the DUT

17.1.2.2.3 Test methodology

17.1.2.2.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

17.1.2.2.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to write a configuration change identified by the URI parameter and className.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 8.1.5.2.

b) The HTTP request is a PATCH operation.

c) The URI matches the configuration being changed.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 8.1.4.2.

Step 4. The test simulator sends the appropriate HTTP response as specified in R1AP [4], clause 8.1.4.2.1.

17.1.2.2.4 Expected result

The test is considered passed if Step 3 validation has passed.

17.1.3 Configuration schema information API as API Consumer test scenario

17.1.3.1 Get all schema information (positive case)

17.1.3.1.1 Test description and applicability

The purpose of this test case is to test the Get all schema information in the Configuration schema information API as specified in R1AP [4], clause 8.3.4.1. The expected outcome is successful validation of the request from the DUT.

17.1.3.1.2 Test entrance criteria

1) The DUT has functionality to initiate the Get schemas procedure as defined in R1GAP [1], clause 5.4.5.
2) A set of schemas exists in the test simulator.

17.1.3.1.3 Test methodology

17.1.3.1.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

17.1.3.1.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to get all schema information for a set of schemas.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 8.3.5.2.
b) The HTTP request is a GET operation.
c) The HTTP request message content includes the information as specified in R1AP [4], clause 8.3.4.1.1.

Step 4. The test simulator generates and sends the appropriate HTTP response as specified in R1AP [4], clause 8.3.4.1.1

17.1.3.1.4 Expected result

The test is considered passed if Step 3 validation has passed.

17.1.3.2 Get individual schema information (positive case)

17.1.3.2.1 Test description and applicability

The purpose of this test case is to test the Get individual schema information in the Configuration schema information API as specified in R1AP [4], clause 8.3.4.2. The expected outcome is successful validation of the request from the DUT.

17.1.3.2.2 Test entrance criteria

1) The DUT has functionality to initiate the Get schemas procedure as defined in R1GAP [1], clause 5.4.5.
2) At least one schema exists in the test simulator and the schemaId is known to the DUT

17.1.3.2.3 Test methodology

17.1.3.2.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

17.1.3.2.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to get schema information for a schema.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 8.3.5.3.

b) The HTTP request is a GET operation.

c) The schemaId in the URI matches the schema information being requested.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 8.3.4.2.1.

Step 4. The test simulator generates and sends the appropriate HTTP response as specified in R1AP [4], clause 8.3.4.2.1.

17.1.3.2.4 Expected result

The test is considered passed if Step 3 validation has passed.

17.2 Conformance test cases for SMO/Non-RT RIC framework

17.2.1 General

17.2.1.1 Device under test requirements

The SMO/Non-RT RIC framework that acts as DUT in the test scenarios, the requirements on the DUT for these tests are that it can handle the RAN OAM Configuration management, and the purpose of the test scenarios is to validate that it conforms to the API Producer functionality as specified in R1AP [4] clause 8.1.4.

17.2.1.2 Test simulator capabilities

The test simulator has the capabilities as required for a rApp. In addition, it has the following capabilities:

- Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.
- Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT's conformance to the API Producer functionality as specified in R1AP **Error! Reference source not found.** clause 8.1.4.

17.2.2 Configuration management as API Producer test scenario

17.2.2.1 Read configuration data (positive case)

17.2.2.1.1 Test description and applicability

The purpose of this test case is to test the read configuration data API as specified in R1AP **Error! Reference source not found.** clause 8.1.4.1. The expected outcome is successful validation of the response from the DUT.

17.2.2.1.2 Test entrance criteria

1) The test simulator has functionality to initiate the Read configuration procedure as defined in R1GAP **Error! Reference source not found.** clause 5.4.5.

2) A managed entity with configuration data exists in the test simulator and the URI parameter and className are known to the test simulator

17.2.2.1.3 Test methodology

17.2.2.1.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

17.2.2.1.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to read configuration data.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The HTTP request is a GET request.

b) The URI conforms to the format specified in R1AP **Error! Reference source not found.** clause 8.1.5.2.

c) The HTTP request message content includes the information as specified in R1AP **Error! Reference source not found.** clause 8.1.4.1.

Step 4. The DUT sends the appropriate HTTP response (200 OK) with configuration data as specified in R1AP [4] clause 8.1.4.1.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The HTTP response message content includes the configuration data as specified in R1AP [4] clause 8.1.4.1 and Clause 8.1.8 (Data model).

b) The response URI conforms to the format specified in R1AP [4] clause 8.1.5.2.

17.2.2.1.4 Expected result

The test is considered passed if Step 6 validations succeed.

17.2.2.2 Write configuration changes (positive case)

17.2.2.2.1 Test description and applicability

The purpose of this test case is to test the Write configuration changes API as specified in R1AP **Error! Reference source not found.** clause 8.1.4.2. The expected outcome is successful validation of the response from the DUT.

17.2.2.2.2 Test entrance criteria

1) The test simulator has functionality to initiate the write configuration procedure as defined in R1GAP **Error! Reference source not found.** clause 5.4.5.

2) A managed entity with configuration data exists in the test simulator and the URI parameter and className are known to the test simulator.

17.2.2.2.3 Test methodology

17.2.2.2.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

17.2.2.2.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to write a configuration change identified by the URI parameter and className.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP **Error! Reference source not found.** clause 8.1.5.2.3.2.

b) The HTTP request is a PATCH operation.

c) The URI Parameter and className in the URI match the configuration being changed.

d) The HTTP request message content includes the information as specified in R1AP **Error! Reference source not found.** clause 8.1.4.2 and Clause 8.1.8 for Data model.

Step 4. The DUT sends the appropriate HTTP response (200 OK or 204 No Content) as specified in R1AP [4] clause 8.1.4.2.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

c) The response URI conforms to the format specified in R1AP [4] clause 8.1.5.2.

d) The HTTP response message content includes the information as specified in R1AP **Error! Reference source not found.** clause 8.1.4.2.

17.2.2.2.4 Expected result

The test is considered passed if Step 6 validations succeed.

17.2.3 Configuration schema information API as API Producer test scenario

17.2.3.1 Get all schema information (positive case)

17.2.3.1.1 Test description and applicability

The purpose of this test case is to test the Get all schema information in the Configuration schema information API clause 8.3.4.1. The expected outcome is successful validation of the request from the DUT.

17.2.3.1.2 Test entrance criteria

1) The test simulator has functionality to initiate the Get schemas procedure as defined in R1GAP [1], clause 5.4.5.
2) A set of schemas exists in the DUT.

17.2.3.1.3 Test methodology

17.2.3.1.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

17.2.3.1.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to get all schema information for a set of schemas.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 8.3.5.2.
b) The HTTP request is a GET operation.
c) The HTTP request message content includes the information as specified in R1AP [4], clause 8.3.4.1.1.

Step 4. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 8.3.4.1.1

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The HTTP response message content includes the information as specified in R1AP [4], clause 8.3.4.1.1.

17.2.3.1.4 Expected result

The test is considered passed if Step 6 validation has passed.

17.3.1.2 Get individual schema information (positive case)

17.3.1.2.1 Test description and applicability

The purpose of this test case is to test the Get individual schema information in the Configuration schema information API clause 8.3.4.2. The expected outcome is successful validation of the request from the DUT.

17.3.1.2.2 Test entrance criteria

1) The test simulator has functionality to initiate the Get schemas procedure as defined in R1GAP [1], clause 5.4.5.
2) At least one schema exists in the DUT and the schemaId is known to the test simulator

17.3.1.2.3 Test methodology

17.3.1.2.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

17.3.1.2.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to get schema information for a schema.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 8.3.5.3.

b) The HTTP request is a GET operation.

c) The schemaId in the URI matches the schema information being requested.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 8.3.4.2.1.

Step 4. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 8.3.4.2.1.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The HTTP response message content includes the information as specified in R1AP [4], clause 8.3.4.2.1.

17.3.1.2.4 Expected result

The test is considered passed if Step 6 validation has passed.

18 RAN OAM Fault management service test cases

18.1 Conformance test cases for rApp

18.1.1 General

18.1.1.1 Device under test requirements

The rApp that acts as DUT in these test scenarios, the requirements on the DUT for these tests are that it can handle the RAN OAM Configuration management service, and the purpose of the test scenarios is to validate that it conforms to the API Consumer functionality as specified in R1AP [4], clause 8.2.4. 18.1.1.2 Test simulator capabilities

The test simulator has the capabilities as required for a SMO/Non-RT RIC framework. In addition, it has the following capabilities:

- Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.

- Controlled initiation of procedures with configurable URIs and payload formulated and modified.

- Validating messages and issuing verdicts related to the procedures in the test cases and thereby enabling determination of the DUT’s conformance to the API Consumer functionality as specified in R1AP [4] clause 8.2.4.

18.1.2 Fault management API as API Consumer test scenario

18.1.2.1 Query alarms (positive case)

18.1.2.1.1 Test description and applicability

The purpose of this test case is to test the Query alarms API as specified in R1AP [4] clause 8.2.4.1. The expected outcome is successful validation of the request from the DUT.

18.1.2.1.2 Test entrance criteria

1) The DUT has functionality to initiate the Query alarms procedure as defined in R1GAP [1] clause 5.4.3.
2) A list of alarms exists in the test simulator and the URI parameter, className and alarmListId are known to the DUT.

18.1.2.1.3 Test methodology

18.1.2.1.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

18.1.2.1.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to query alarms.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4] clause 8.2.5.2.

b) The HTTP request is a GET operation.

c) The HTTP request message content includes the information as specified in R1AP [4] clause 8.2.5.2.

Step 4. The test simulator constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4] clause 8.2.4.1.1.

18.1.2.1.4 Expected result

The test is considered passed if Step 3 validation has passed. 18.1.2.2 Change alarm acknowledgement state (positive case)

18.1.2.2.1 Test description and applicability

The purpose of this test case is to test the Change alarm acknowledgement state API as specified in R1AP [4] clause 8.2.4.2. The expected outcome is successful validation of the request from the DUT.

18.1.2.2.2 Test entrance criteria

1) The DUT has functionality to initiate the Change alarm acknowledgement state procedure as defined in R1GAP [1] clause 5.4.3.
2) A list of alarms exists in the test simulator and the URI parameter, className and alarmListId are known to the DUT.

18.1.2.2.3 Test methodology

18.1.2.2.3.1 Initial conditions

1) The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.
2) The alarm list identifier is known to the test simulator and the resource URI either "SubNetwork" or "ManagedElement" as the class name, the related object instance identifier for the alarmListId is known to DUT.

18.1.2.2.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to change the alarm acknowledgement state identified by the URI parameter, className and alarmListId.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4] clause 8.2.5.2.

b) The HTTP request is a PATCH operation.

c) The URI Parameter and className in the URI match the alarmListId alarm acknowledgement state being changed.

d) The HTTP request message content includes the information as specified in R1AP [4] clause 8.2.5.2.

Step 4. The test simulator constructs the URI for the created resource and sends the appropriate HTTP response as specified in R1AP [4] clause 8.2.5.2.

18.1.2.2.4 Expected result

The test is considered passed if Step 3 validation has passed.

18.2 Conformance test cases for SMO/Non-RT RIC framework

18.2.1 General

18.2.1.1 Device under test requirements

The SMO/Non-RT RIC framework that acts as DUT in the test scenarios, the requirements on the DUT for these tests are that it can handle the Fault management service, and the purpose of the test scenarios is to validate that it conforms to the API Producer functionality as specified in R1AP [4] clause 8.2.4.

18.2.1.2 Test simulator capabilities

The test simulator has the capabilities as required for a rApp. In addition, it has the following capabilities:

- Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.
- Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT's conformance to the API Producer functionality as specified in R1AP **Error! Reference source not found.** clause 8.2.4.

18.2.2 Fault management as API Producer test scenario

18.2.2.1 Query alarms (positive case)

18.2.2.1.1 Test description and applicability

The purpose of this test case is to test the Query alarms API as specified in R1AP **Error! Reference source not found.** clause 8.2.4.1. The expected outcome is successful validation of the response from the DUT.

18.2.2.1.2 Test entrance criteria

1) The test simulator has functionality to initiate the Query alarms procedure as defined in R1GAP **Error! Reference source not found.** clause 5.4.3.
2) A list of alarms exists in the test simulator and the URI parameter, className and alarmListId are known to the test simulator

18.2.2.1.3 Test methodology

18.2.2.1.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

18.2.2.1.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to query alarms.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The HTTP request is a GET request.
b) The URI conforms to the format specified in R1AP **Error! Reference source not found.** clause 8.2.5.2.3.1.
c) The HTTP request message content includes the information as specified in R1AP **Error! Reference source not found.** clause 8.2.5.2.3.1

Step 4. The DUT sends the appropriate HTTP response (200 OK) with the alarm list as specified in R1AP **Error! Reference source not found.** clause 8.2.5.2.3.1

Step 5. At the test simulator the contents of the received HTTP response.

Step 6. The test simulator does the following validation:

a) The HTTP response code is 200 OK
b) The HTTP response message content includes the information as specified in R1AP **Error! Reference source not found.**, clause 8.2.5.2.3.1 and clause 8.2.8 for Data model.

18.2.2.1.4 Expected result

The test is considered passed if Step 6 validations succeed.

18.2.2.2 Change alarm acknowledgement state (positive case)

18.2.2.2.1 Test description and applicability

The purpose of this test case is to test the Change alarm acknowledgement state API as specified in R1AP **Error! Reference source not found.** clause 8.2.4.2. The expected outcome is successful validation of the request from the DUT.

18.2.2.2.2 Test entrance criteria

1) The test simulator has functionality to initiate the Change alarm acknowledgement state procedure as defined in R1GAP **Error! Reference source not found.** clause 5.4.3.

2) A list of alarms exists in the test simulator and the URI parameter, className and alarmListId are known to the test simulator.

18.2.2.2.3 Test methodology

18.2.2.2.3.1 Initial conditions

1. The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.
2. The alarm list identifier is known to the DUT and the resource URI (either “SubNetwork“ or “ManagedElement“ as the class name) and the related object instance identifier for the alarmListId is known to test simulator.

18.2.2.2.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to change the alarm acknowledgement state identified by the URI parameter, className and alarmListId.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP **Error! Reference source not found.** clause 8.2.5.2.3.2.

b) The HTTP request is a PATCH operation.

c) The URI Parameter and className and alarmListId, match the alarm acknowledgement state being changed.

d) The HTTP request message content includes the information as specified in R1AP **Error! Reference source not found.** clause 8.2.5.2.3.2.

Step 4. The DUT sends the appropriate HTTP response (200 OK or 204 No Content) as specified in R1AP **Error! Reference source not found.** clause 8.2.5.2.3.2.

Step 5. At the test simulator the contents of the received HTTP response.

Step 6. The test simulator does the following validation:

a) The HTTP response code is 200 OK or 204 No Content.

b) The HTTP response message content includes the information as specified in R1AP **Error! Reference source not found.** clause 8.2.5.2.3.2 and clause 8.2.8 for Data model.

18.2.2.2.4 Expected result

The test is considered passed if Step 6 validation has passed.

19 DME Data delivery service test cases

19.1 Conformance test cases for rApp

19.1.1 General

19.1.1.1 Device under test requirements

The rApp acts as DUT in these test scenarios, the requirements on the DUT for these tests are that it can handle the DME Data delivery service, and the purpose of the test scenarios is to validate that it conforms to the API Producer functionality as specified in R1AP [4], clause 7.4.4.

19.1.1.2 Test simulator capabilities

The test simulator has the capabilities as required for a SMO/Non-RT RIC framework. In addition, it has the following capabilities:

- Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.
- Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT's conformance to the API Producer functionality as specified in R1AP [4], clause 7.4.4.

19.1.2 HTTP based Push data API as API producer test scenario

19.1.2.1 Push data (positive case)

19.1.2.1.1 Test description and applicability

The purpose of this test case is to test the Push data in the HTTP based Push data API as specified in R1AP [4], clause 7.4.4.1. The expected outcome is successful validation of the request from the DUT.

19.1.2.1.2 Test entrance criteria

The DUT has functionality to initiate the Push data procedure as defined in R1GAP [1], clause 5.2.6.

19.1.2.1.3 Test methodology

19.1.2.1.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP requests from the DUT.

19.1.2.1.3.2 Procedure

Step 1. The DUT as an API Producer initiates a HTTP request to push a data payload.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.4.5.2.2.

b) The HTTP request is a POST operation.

c) The HTTP request message content includes the information as specified in R1AP [4], clause 7.4.4.1.1.

Step 4. The test simulator generates and sends the appropriate HTTP response as specified in R1AP [4], clause 7.4.4.1.1.

19.1.2.1.4 Expected result

The test is considered passed if Step 3 validation has passed.

19.1.3 HTTP based Pull data API as API Consumer test scenario

19.1.3.1 Pull data (positive case)

19.1.3.1.1 Test description and applicability

The purpose of this test case is to test the Pull data in the HTTP based Pull data API as specified in R1AP [4], clause 7.5.4.1. The expected outcome is successful validation of the request from the DUT.

19.1.3.1.2 Test entrance criteria

The DUT has functionality to initiate the Pull data procedure as defined in R1GAP [1], clause 5.2.6.

19.1.3.1.3 Test methodology

19.1.3.1.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

19.1.3.1.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to pull a data payload.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.5.5.2.2.

b) The HTTP request is a GET operation.

c) The HTTP request message content includes the information as specified in R1AP [4], clause 7.5.4.1.1.

Step 4. The test simulator generates and sends the appropriate HTTP response as specified in R1AP [4], clause 7.5.4.1.1.

19.1.3.1.4 Expected result

The test is considered passed if Step 3 validation has passed.

19.2 Conformance test cases for SMO/Non-RT RIC framework

19.2.1 General

19.2.1.1 Device under test requirements

The SMO/Non-RT RIC framework that acts as DUT in these test scenarios, the requirements on the DUT for these tests are that it can handle the DME Data delivery service, and the purpose of the test scenarios is to validate that it conforms to the API Producer functionality as specified in R1AP [4], clause 7.4.4.

19.2.1.2 Test simulator capabilities

The test simulator has the capabilities as required for a rApp. In addition, it has the following capabilities:

- Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.
- Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT's conformance to the API Producer functionality as specified in R1AP [4], clause 7.4.4.

19.2.2 HTTP based Push data API as API Consumer test scenario

19.2.2.1 Push data (positive case)

19.2.2.1.1 Test description and applicability

The purpose of this test case is to test the Push data in the HTTP based Push data API as specified in R1AP [4], clause 7.4.4.1. The expected outcome is successful validation of the request from the DUT.

19.2.2.1.2 Test entrance criteria

The DUT has functionality to initiate the Push data procedure as defined in R1GAP [1], clause 5.2.6.

19.2.2.1.3 Test methodology

19.2.2.1.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

19.2.2.1.3.2 Procedure

Step 1. The DUT as an API Producer initiates a HTTP request to push a data payload.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.4.5.2.2.
b) The HTTP request is a POST operation.
c) The HTTP request message content includes the information as specified in R1AP [4], clause 7.4.4.1.1.

Step 4. The test simulator sends the appropriate HTTP response as specified in R1AP [4], clause 7.4.4.1.1.

19.2.2.1.4 Expected result

The test is considered passed if Step 3 validation has passed.

19.2.3 HTTP based Pull data API as API Producer test scenario

19.2.3.1 Pull data (positive case)

19.2.3.1.1 Test description and applicability

The purpose of this test case is to test the Pull data in the HTTP based Pull data API as specified in R1AP [4], clause 7.5.4.1. The expected outcome is successful validation of the response from the DUT.

19.2.3.1.2 Test entrance criteria

The test simulator has functionality to initiate the Pull data procedure as defined in R1GAP [1], clause 5.2.6.

19.2.3.1.3 Test methodology

19.2.3.1.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

19.2.3.1.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to pull a data payload.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 7.5.5.2.2.

b) The HTTP request is a GET operation.

c) The HTTP request message content includes the information as specified in R1AP [4], clause 7.5.4.1.1.

Step 4. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 7.5.4.1.1.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The HTTP response message content includes the information as specified in R1AP [4], clause 7.5.4.1.1.

19.2.3.1.4 Expected result

The test is considered passed if Step 6 validation has passed

20 AI/ML workflow AI/ML model training API test cases

20.1 Conformance test cases for rApp

20.1.1 General

20.1.1.1 Device under test requirements

The rApp that acts as DUT in these test scenarios, the requirements on the DUT for these tests are that it can handle the AI/ML training services, and the purpose of the test scenarios is to validate that it conforms to the API Consumer functionality as specified in R1AP [4], clause 10.3.4.

20.1.1.2 Test simulator capabilities

The test simulator has the capabilities as required for a SMO/Non-RT RIC framework. In addition, it has the following capabilities:

- Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.
- Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT's conformance to the API Consumer functionality as specified in R1AP [4], clause 10.3.4.

20.1.2 AI/ML model training API as API Consumer test scenario

20.1.2.1 Request AI/ML model training (positive case)

20.1.2.1.1 Test description and applicability

The purpose of this test case is to test the request AI/ML model training as specified in R1AP [4], clause 10.3.4.1. The expected outcome is successful validation of the request from the DUT.

20.1.2.1.2 Test entrance criteria

The DUT has functionality to initiate the Request AI/ML model training procedure as defined in R1GAP [1], clause 5.6.8.

20.1.2.1.3 Test methodology

20.1.2.1.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

20.1.2.1.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request for AI/ML model training.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 10.3.5.2.2.

b) The HTTP request is a POST operation.

c) The HTTP request message content includes the information as specified in R1AP [4], clause 10.3.4.1.1.

Step 4. The test simulator sent the appropriate HTTP response as specified in R1AP [4], clause 10.3.4.1.1.

20.1.2.1.4 Expected result

The test is considered passed if Step 3 validation has passed.

20.1.2.2 Cancel AI/ML model training (positive case)

20.1.2.2.1 Test description and applicability

The purpose of this test case is to test the cancel AI/ML model training as specified in R1AP [4], clause 10.3.4.2. The expected outcome is successful validation of the request from the DUT.

20.1.2.2.2 Test entrance criteria

1) The DUT has functionality to initiate the Cancel AI/ML model training procedure as defined in R1GAP [1], clause 5.6.8.

2) A training job exists in the test simulator and the trainingJobId is known to the DUT.

20.1.2.2.3 Test methodology

20.1.2.2.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

20.1.2.2.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP to cancel AI/ML model training.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 10.3.5.3.2.

b) The HTTP request is a DELETE operation.

c) The HTTP request message content includes the information as specified in R1AP [4], clause 10.3.4.2.1.

Step 4. The test simulator sent the appropriate HTTP response as specified in R1AP [4], clause 10.3.4.2.1.

20.1.2.2.4 Expected result

The test is considered passed if Step 3 validation has passed.

20.1.2.3 Query AI/ML model training job status (positive case)

20.1.2.3.1 Test description and applicability

The purpose of this test case is to test the Query AI/ML model training job status as specified in R1AP [4], clause 10.3.4.3. The expected outcome is successful validation of the request from the DUT.

20.1.2.3.2 Test entrance criteria

1) The DUT has functionality to initiate the Query AI/ML model training job status procedure as defined in R1GAP [1], clause 5.6.8.

2) A training job exists in the test simulator and the trainingJobId is known to the DUT.

20.1.2.3.3 Test methodology

20.1.2.3.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

20.1.2.3.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request to Query a training job status.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 10.3.5.4.2.

b) The HTTP request is a GET operation.

c) The trainingJobId in the URI match the training job status being queried.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 10.3.4.3.1.

Step 4. The test simulator sent the appropriate HTTP response as specified in R1AP [4], clause 10.3.4.3.1.

20.1.2.3.4 Expected result

The test is considered passed if Step 3 validation has passed.

20.2 Conformance test cases for SMO/Non-RT RIC framework

20.2.1 General

20.2.1.1 Device under test requirements

The SMO/Non-RT RIC framework that acts as Device Under Test (DUT) in these test scenarios, the requirements on the DUT for these tests are that it can handle the AI/ML model training services, and the purpose of the test scenarios is to validate that it conforms to the AI/ML model training API specified in R1AP [4], clause 10.3.4.

20.2.1.2 Test simulator capabilities

The test simulator has the capabilities as specified in clause 4.3.2. In addition, it has the following capabilities:

1) Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.

2) Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT’s conformance to API Producer functionality as specified in R1AP [4], clause 10.3.4.

20.2.2 AI/ML model training API as API Producer test scenario

20.2.2.1 Request AI/ML model training (positive case)

20.2.2.1.1 Test description and applicability

The purpose of this test case is to test the Request AI/ML model training APIs as specified in R1AP [4], clause 10.3.4.1. The expected outcome is successful validation of the response from the DUT.

20.2.2.1.2 Test entrance criteria

The test simulator supports the functionality to initiate the Request AI/ML model training procedure as defined in R1GAP [1], clause 5.6.8.

20.2.2.1.3 Test methodology

20.2.2.1.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

20.2.2.1.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request for AI/ML model training.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The HTTP request is a POST request.

b) The URI conforms to the format as specified in R1AP [4], clause 10.3.5.2.2

c) The HTTP request message content includes the information as specified in R1AP [4], clause 10.3.4.1.1.

Step 4. The DUT initiates a HTTP Response.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 10.3.5.2.2.

b) The HTTP response message content includes the information as specified in R1AP [4], clause 10.3.4.1.1.

20.2.2.1.4 Expected result

The test is considered passed if Step 6 validation has passed.

20.2.2.2 Cancel AI/ML model training (positive case)

20.2.2.2.1 Test description and applicability

The purpose of this test case is to test the Cancel AI/ML model training APIs as specified in R1AP [4], clause 10.3.4.2. The expected outcome is successful validation of the response from the DUT.

20.2.2.2.2 Test entrance criteria

The test simulator supports the functionality to initiate the Cancel AI/ML model training procedure as defined in R1GAP [1], clause 5.6.8.

A training job exists in the DUT and the trainingJobId is known to the test simulator.

20.2.2.2.3 Test methodology

20.2.2.2.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

20.2.2.2.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to cancel AI/ML model training.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 10.3.5.3.2.

b) The HTTP request is a DELETE operation.
c) The trainingJobId in the URI match the training job being cancelled.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 10.3.4.2.1.

Step 4. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 10.3.4.2.1.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 10.3.5.3.2.

b) The HTTP response message content includes the information as specified in R1AP [4], clause 10.3.4.2.1.

20.2.2.2.4 Expected result

The test is considered passed if Step 6 validation has passed.

20.2.2.3 Query AI/ML model training job status (positive case)

20.2.2.3.1 Test description and applicability

The purpose of this test case is to test the Query AI/ML model training job status as specified in R1AP [4], clause 10.3.4.3. The expected outcome is successful validation of the request from the DUT.

20.2.2.3.2 Test entrance criteria

1) The test simulator has functionality to initiate the Query AI/ML model training job status procedure as defined in R1GAP [1], clause 5.6.8.

2) A training job exists in the DUT and the trainingJobId is known to the test simulator.

20.2.2.3.3 Test methodology

20.2.2.3.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

20.2.2.3.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request to query a training job status.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 10.3.5.4.2.

b) The HTTP request is a GET operation.

c) The trainingJobId in the URI match the training job status being queried.

d) The HTTP request message content includes the information as specified in R1AP [4], clause 10.3.4.3.1.

Step 4. The DUT sends the appropriate HTTP response as specified in R1AP [4], clause 10.3.4.3.1.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 10.3.5.4.2.

b) The HTTP response message content includes the information as specified in R1AP [4], clause 10.3.4.3.1.

20.2.2.3.4 Expected result

The test is considered passed if Step 6 validation has passed.

20.2.2.4 Notify AI/ML model training job status change (positive case)

20.2.2.4.1 Test description and applicability

The purpose of this test case is to test the Notify AI/ML model training job status change APIs as specified in R1AP [4], clause 10.3.4.4. The expected outcome is successful validation of the response from the DUT.

20.2.2.4.2 Test entrance criteria

1) The DUT supports the functionality to initiate the Notify AI/ML model training job status change procedure as defined in R1GAP [1], clause 5.6.8.

2) A training job exists in the DUT.

20.2.2.4.3 Test methodology

20.2.2.4.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP requests from the DUT.

20.2.2.4.3.2 Procedure

Step 1. The DUT as an API producer initiates the sending of a HTTP POST request.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The HTTP request is a POST operation.

b) The callback URI matches the URI provided in the creation of the training job.

Step 4. The test simulator sends the HTTP response as specified in R1AP [4], clause 10.3.4.4.1.

20.2.2.4.4 Expected result

The test is considered passed if Step 3 validation has passed.

21 AI/ML workflow AI/ML model deployment API test cases

21.1 Conformance test cases for rApp

21.1.1 General

21.1.1.1 Device under test requirements

The rApp that acts as DUT in these test scenarios, the requirements on the DUT for these tests are that it can handle the AI/ML model deployment request services, and the purpose of the test scenarios is to validate that it conforms to the API Consumer functionality as specified in R1AP [4], clause 10.4.4.

21.1.1.2 Test simulator capabilities

The test simulator has the capabilities as required for a SMO/Non-RT RIC framework. In addition, it has the following capabilities:

- Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.
- Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT's conformance to the API Consumer functionality as specified in R1AP [4], clause 10.4.4.

21.1.2 AI/ML model deployment API as API Consumer test scenario

21.1.2.1 Request AI/ML model deployment (positive case)

21.1.2.1.1 Test description and applicability

The purpose of this test case is to test the request AI/ML model deployment as specified in R1AP [4], clause 10.4.4.1. The expected outcome is successful validation of the request from the DUT.

21.1.2.1.2 Test entrance criteria

The DUT has functionality to initiate the Request AI/ML model deployment procedure as defined in R1GAP [1], clause 5.6.7.

21.1.2.1.3 Test methodology

21.1.2.1.3.1 Initial conditions

The test simulator as API Producer is ready and available to receive HTTP requests from the DUT.

21.1.2.1.3.2 Procedure

Step 1. The DUT as an API Consumer initiates a HTTP request for AI/ML model deployment.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The URI confirms to the format specified in R1AP [4], clause 10.4.5.2.

b) The HTTP request is a POST operation.

c) The HTTP request message content includes the information as specified in R1AP [4], clause 10.4.4.1.1.

Step 4. The test simulator sent the appropriate HTTP response as specified in R1AP [4], clause 10.4.4.1.1.

21.1.2.1.4 Expected result

The test is considered passed if Step 3 validation has passed.

21.2 Conformance test cases for SMO/Non-RT RIC framework

21.2.1 General

21.2.1.1 Device under test requirements

The SMO/Non-RT RIC framework that acts as Device Under Test (DUT) in these test scenarios, the requirements on the DUT for these tests are that it can handle the AI/ML model deployment request services, and the purpose of the test scenarios is to validate that it conforms to the AI/ML model deployment API specified in R1AP [4], clause 10.4.4.

21.2.1.2 Test simulator capabilities

The test simulator has the capabilities as specified in clause 4.3.2. In addition, it has the following capabilities:

1) Recording of received HTTP requests and responses and analyzing them regarding conformance to the R1 service definitions.

2) Controlled initiation of procedures with configurable URIs and payload formulated and modified.

Validating messages and issuing of verdicts related to the procedures in the test cases and thereby enabling determination of the DUT's conformance to API Producer functionality as specified in R1AP [4], clause 10.4.4.

21.2.2 AI/ML model deployment API as API Producer test scenario

21.2.2.1 Request AI/ML model deployment (positive case)

21.2.2.1.1 Test description and applicability

The purpose of this test case is to test the Request AI/ML model deployment APIs as specified in R1AP [4], clause 10.4.4.1. The expected outcome is successful validation of the response from the DUT.

21.2.2.1.2 Test entrance criteria

The test simulator supports the functionality to initiate the Request AI/ML model deployment procedure as defined in R1GAP [1], clause 5.6.7.

21.2.2.1.3 Test methodology

21.2.2.1.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP responses from the DUT.

21.2.2.1.3.2 Procedure

Step 1. The test simulator as an API Consumer initiates a HTTP request for AI/ML model deployment.

Step 2. The DUT receives the HTTP request.

Step 3. The DUT does the following validation:

a) The HTTP request is a POST request.

b) The URI conforms to the format as specified in R1AP [4], clause 10.4.5.2

c) The HTTP request message content includes the information as specified in R1AP [4], clause 10.4.4.1.1.

Step 4. The DUT initiates a HTTP Response.

Step 5. At the test simulator the contents of the received HTTP response is recorded.

Step 6. The test simulator does the following validation:

a) The URI conforms to the format specified in R1AP [4], clause 10.4.5.2.

b) The HTTP response message content includes the information as specified in R1AP [4], clause 10.4.4.1.1.

21.2.2.1.4 Expected result

The test is considered passed if Step 6 validation has passed.

21.2.2.2 Notify AI/ML model deployment status change (positive case)

21.2.2.2.1 Test description and applicability

The purpose of this test case is to test the Notify AI/ML model deployment job status change APIs as specified in R1AP [4], clause 10.4.4.2. The expected outcome is successful validation of the response from the DUT.

21.2.2.2.2 Test entrance criteria

1) The DUT supports the functionality to initiate the Request AI/ML model deployment procedure as defined in R1GAP [1], clause 5.6.7

2) A model exists in the DUT and the notificationDestination is known to the test simulator.

21.2.2.2.3 Test methodology

21.2.2.2.3.1 Initial conditions

The test simulator as API Consumer is ready and available to receive HTTP requests from the DUT.

21.2.2.2.3.2 Procedure

Step 1. The DUT as an API producer initiates the sending of a HTTP POST request.

Step 2. At the test simulator the contents of the received HTTP request is recorded.

Step 3. The test simulator does the following validation:

a) The HTTP request is a POST operation.

b) The callback URI matches the URI provided in the model deployment request.

Step 4. The test simulator sends the HTTP response as specified in R1AP [4], clause 10.4.4.2.1.

21.2.2.2.4 Expected result

The test is considered passed if Step 3 validation has passed.

Annex (informative): Change history

Date	Revision	Description
2025.11.24	05.00	Included The test cases related to Configuration management service, Fault management Service, DME -Query Data subscription, HTTP PUSH , HTTP Pull, Config schema , AI/ML model training, AI/ML model deployment, AI/ML model discovery.
2025.07.10	04.00	Published with conformance test of rApp and Non-RT RIC as DUT for Data offer testcases, Boot strap test case, AI/ML model discovery testcase , AI/ML model registration testcase and Configuration management testcase for rApp as DUT.
2025.03.13	03.00	Published with Conformance test of rApp and Non-RT RIC as DUT for DME, A1 policy management and SME
2024.11.21	02.00	Published with Conformance test of Non-RT RIC as DUT for Service registration test case
2024.07.11	01.00	Published with conformance testing of rApps and SMO/Non-RT RIC Framework and with one Service registration test case

